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You the expert in CFA level of Knowledge , I will provide you some question related to Derivatives topic , i want you to answer with your best level of knowledge.

Absolutely. I'll handle your **CFA Level I Derivatives** questions at an exam-focused level.

For each question, I'll give you:

1. **Correct answer** — A/B/C/D
2. **Why it is correct** — explained from the fundamentals
3. **Why the other choices are wrong**
4. **CFA exam rule/concept** to remember
5. **Simple numerical example** when it helps
6. **Exam shortcut / trap** — especially useful for distinguishing similar choices

I'll be careful with the CFA terminology around **forwards, futures, options, swaps, hedging, arbitrage, credit risk, settlement, and derivative payoffs**.

Most importantly, if you say **"I did not get this"**, I'll explain it **from scratch in very simple language**, rather than just repeating the textbook explanation.

Send me your first Derivatives question.

Consider the following structured note offered by Baywhite Financial: Baywhite Financial LLC 80% Principal Protected Structured Note Description: The Baywhite Financial LLC 80% Principal Protected Structured Note ("the Note") is linked to the performance of the S&P 500 Health Care Select Sector Index (SIXV). Issuer: Baywhite Financial LLC Start Date: [Today] Maturity Date: [Six

months from Start Date] Issuance Price: 102% of Face Value Face Value: Sold in a minimum denomination of USD1,000 and multiple units thereof Payment at Maturity:

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Got it. This is a **structured note vs. stand-alone derivative** question. Before answering any specific multiple-choice question that follows, let's decode exactly what the Baywhite note is doing.

1. What is the investor actually buying?

The note has two components:

- **80% principal protection**
- **An upside-linked payoff** based on the S&P 500 Health Care Select Sector Index (SIXV)

But there is an important catch:

> The note is **issued at 102% of face value**.

So for a $1,000 face-value note, the investor actually pays:

$1,000 × 102% = $1,020

Yet the guaranteed portion at maturity is only:

$1,000 × 80% = $800

Therefore, the investor has **$220 of effective downside from the original investment**, before considering any additional amount.

---

2. How does the Additional Amount work?

The note says the investor receives:

> the greater of **100% of the SIXV return in excess of 5%**, or zero.

So there is effectively a **5% threshold**.

Let:

$$R = \text{return on SIXV}$$

Then:

$$\text{Additional Amount} = \max[1000(R - 5\%), 0]$$

Therefore:

$$\boxed{\text{Maturity Payment} = 800 + \max[1000(R - 5\%), 0]}$$

Notice something very important:

The investor does NOT participate in the first 5% of index gains.

For example:

SIXV return	Additional Amount	Maturity payment
−20%	$0	$800
0%	$0	$800
+5%	$0	$800
+8%	$30	$830
+15%	$100	$900
+30%	$250	$1,050

But remember, the investor paid **$1,020**, not $1,000.

So even if the index rises by 20%, the investor receives:

$$800 + (20\% - 5\%)1000 = \$950$$

That's still a **loss of $70** relative to the $1,020 actually invested.

---

3. The key CFA concept: decompose the structured note

This is where the question is likely going.

A structured note can generally be viewed as a combination of:

$$\boxed{\text{Structured Note} = \text{Debt Instrument} + \text{Derivative(s)}}$$

For this particular note, conceptually:

Debt component

The issuer promises at least **$800 at maturity**.

So there is a debt-like component providing the 80% principal protection.

Derivative component

The additional payment depends on the SIXV's performance.

The payoff:

$$\max[1000(R-5\%),0]$$

has the basic shape of an **option payoff**.

The investor benefits from upside in the index **above a 5% threshold**, while receiving nothing from the derivative if the index return is ≤ 5%.

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4. What does "20% principal at risk" really mean?

This is a common CFA trap.

80% principal protection does NOT mean the investor can only lose 20% of what they paid.

Why?

Because the note is issued at **102% of face value**.

Investor pays:

$$\$1,020$$

Minimum maturity payment:

$$\$800$$

Potential loss:

$$1020-800=\boxed{\$220}$$

Percentage loss relative to the actual investment:

$$\frac{220}{1020}\approx\boxed{21.57\%}$$

So the **80% principal protection is based on face value**, not the investor's actual purchase price.

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5. Another major risk: issuer credit risk

The $800 "protection" is a promise by **Baywhite Financial LLC**.

It is therefore **not the same thing as a government-guaranteed principal protection**.

If Baywhite defaults, the investor may not receive the promised $800.

This is an important distinction between a structured note and a stand-alone exchange-traded derivative.

🧠 CFA exam takeaway

When you see a question asking you to compare this note with a stand-alone derivative, immediately think:

Structured Note

↓

Debt obligation of issuer
+
Embedded derivative linked to SIXV

And remember these four things:

$\boxed{\text{80\% protection is based on face value}}$
$\boxed{\text{Investor pays 102\% of face value}}$

$$\boxed{\text{No upside participation until SIXV gains } >5\%}$$

$$\boxed{\text{Issuer credit risk exists}}$$

If you provide the **multiple-choice question that follows this case**, I'll walk through exactly which answer CFA is looking for and why each alternative is right or wrong.

1. Which of the following statements best describes the derivative instrument that
is embedded in the Baywhite Financial LLC Structured Note?
A. The Structured Note has an embedded long futures contract with the S&P

500 Health Care Select Sector Index (SIXV) as an underlying.
B. The Structured Note has an embedded long call option contract with the

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Correct Answer: B. Long call option

Let's break this down from **zero**, because this is a very common CFA Derivatives pattern.

Step 1: Look at the payoff

The note says the investor receives:

> **100% of the return of SIXV above 5%, or zero**

So the additional amount is:

$$\text{Additional Amount} = \max[(R - 5\%), 0]$$

This is the key.

The payoff is **zero when SIXV does not rise above 5%**, and becomes increasingly positive when SIXV rises above 5%.

That is exactly the shape of a **call option**.

Call option payoff

$$\boxed{\text{Call Payoff} = \max(S_T - K, 0)}$$

where:

- S_T = value of underlying at maturity
- K = strike price

Here, the **5% return threshold acts like the strike**.

So:

$$\boxed{\text{Additional Amount} \sim \max(R - 5\%, 0)}$$

Therefore, the investor has a **long call** embedded in the note.

Why B is correct

A **long call** gives the investor:

- **Upside participation** if the underlying increases above the strike.
- **No payoff** if the underlying remains below the strike.

That's exactly what happens here.

Example

Suppose SIXV increases by:

3%

The return does not exceed 5%.

$$\max(3\% - 5\%, 0) = 0$$

Additional amount **= $0**

Now suppose SIXV increases by:

12%

The amount above the 5% threshold is:

$$12\% - 5\% = 7\%$$

So the investor receives the positive payoff associated with that excess return.

That's **long call behavior**.

Why A is wrong — Long futures

A **long futures contract** gives you approximately:

$$S_T - S_0$$

So you participate in the **entire movement** of the underlying.

If SIXV rises 3%, a long futures position benefits from that 3%.

But here:

SIXV +3% → Additional Amount = $0

So this cannot be a long futures contract.

Remember:

Futures = symmetric payoff

- Underlying ↑ → gain
- Underlying ↓ → loss

Call = asymmetric payoff

- Underlying ↑ above strike → gain
- Underlying ≤ strike → zero payoff

The structured note has the **asymmetric call payoff**.

Why C is wrong — Short put

A **short put** benefits when the underlying stays above the strike, but its payoff is:

$$\boxed{-\max(K - S_T, 0)}$$

Its payoff becomes negative when the underlying falls below the strike.

The embedded derivative here does **not** create a negative payoff for the investor.

Instead:

$$\boxed{\text{Payoff} = \max(R - 5\%, 0)}$$

That's a **long call**, not a short put.

🧠 CFA Exam Shortcut

Whenever you see:

> **"Investor receives the greater of [underlying return − threshold] or zero"**

Immediately think:

$$\boxed{\max(S - K, 0) = \text{LONG CALL}}$$

Payoff recognition table

Position	Payoff	What it means
Long call	$\max(S - K, 0)$	Benefits from upside

Position	Payoff	What it means
Short call	$-\max(S-K,0)$	Hurt by upside
Long put	$\max(K-S,0)$	Benefits from downside
Short put	$-\max(K-S,0)$	Hurt by downside
Long futures	S_T-S_0	Symmetric upside/downside

Therefore:

$$\boxed{\textbf{B. Embedded long call option}}$$

The 5% threshold is effectively the call's strike/participation threshold.

I could not undersatnd it completly

Absolutely. Let's forget the complicated structured-note language for a moment and build this **from the very beginning**.

The key is to understand **why this is a long call and not a futures contract or short put**.

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1. First, ignore the $800 and $1,020

For this question, we only care about the **Additional Amount**.

The note says:

> You get the return of SIXV **above 5%**, but if the return is not above 5%, you get **zero**.

So imagine the following:

SIXV return	Amount above 5%	What you get
0%	Nothing	$0
3%	Nothing	$0
5%	Nothing	$0

SIXV return	Amount above 5%	What you get
7%	2%	Positive amount
10%	5%	Positive amount
20%	15%	Positive amount

Look carefully at this pattern.

If SIXV doesn't go above 5%:

You get nothing.

If SIXV goes above 5%:

You start making money.

That is exactly what a **call option** does.

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2. Let's understand a CALL OPTION first

Suppose I give you this option:

> "You have the right to buy a stock at ₹100."

I charge you for this right.

Now imagine the stock price at expiration is:

Case 1: Stock = ₹90

Why would you buy it for ₹100 when you can buy it in the market for ₹90?

So you don't exercise the option.

Your payoff:

$$\boxed{0}$$

Case 2: Stock = ₹110

Now you have a valuable right.

You can buy it for ₹100 and the stock is worth ₹110.

Your gain:

$$110 - 100 = \boxed{₹10}$$

Case 3: Stock = ₹150

You buy for ₹100 something worth ₹150.

Gain:

$$150 - 100 = \boxed{₹50}$$

So the call option behaves like this:

Stock price	Call payoff
₹90	₹0
₹100	₹0
₹110	₹10
₹150	₹50

Notice the pattern:

Below the threshold → ZERO

Above the threshold → POSITIVE payoff

That's the most important thing.

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3. Now compare this with the Baywhite note

Baywhite says:

> "I'll give you the SIXV return **above 5%**."

So let's pretend the 5% threshold is our **₹100 strike price**.

The logic becomes:

SIXV return = 3%

It's below 5%.

Therefore:

$$\boxed{0}$$

SIXV return = 5%

It's exactly at 5%.

Therefore:

$$\boxed{0}$$

SIXV return = 8%

It's 3% above 5%.

Therefore:

$$\boxed{3\%}$$

SIXV return = 15%

It's 10% above 5%.

Therefore:

$$\boxed{10\%}$$

Look at the pattern:

SIXV return	Additional Amount
3%	0
5%	0
8%	3%
15%	10%

This is **exactly the behavior of a call option**.

Therefore:

$$\boxed{\text{Embedded derivative} = \text{Long Call}}$$

---

4. Now let's understand why it's LONG call

This word **LONG** is very important.

When you **buy** a call option, you are **long the call**.

A long call means:

> "I want the underlying to go UP."

If SIXV goes up substantially, the investor benefits.

That's exactly what the note does.

SIXV goes up → investor gets more money

Therefore:

$$\boxed{\text{Investor is LONG the upside}}$$

So the embedded derivative is a:

$$\boxed{\text{LONG CALL}}$$

---

5. Why isn't it a futures contract?

This is probably where the confusion comes from.

A **futures contract** behaves very differently.

Suppose you enter a **long futures** position.

You benefit from **every increase** in the underlying.

There is no "5% free zone."

For example:

SIXV return	Long futures
+2%	Gain
+3%	Gain
+5%	Gain
+10%	Gain

But Baywhite's note:

SIXV return	Baywhite additional amount
+2%	**$0**
+3%	**$0**
+5%	**$0**
+10%	**Positive**

Therefore it **cannot be a futures contract**.

Think:

Futures = participate immediately

Call = participate only after the strike

The Baywhite note participates only after **5%**.

So → **CALL**.

---

6. Why isn't it a short put?

A **short put** is also potentially confusing because it can make money when the underlying goes up or stays above the strike.

But look at the payoff.

A short put has this type of behavior:

Underlying	Short put
Goes up	Profit
Stays around strike	Profit/zero
Goes DOWN significantly	**Loss**

The investor in a short put is taking **downside risk**.

But the additional amount in this note can never be negative:

$$\boxed{\max(R - 5\%, 0)}$$

It is either:

positive OR zero.

Never negative.

That's the signature of a **long call**.

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7. The easiest way to remember this for CFA

Forget the complicated wording.

Look for these patterns.

Pattern 1

$$\boxed{\max(S - K, 0)}$$

Underlying goes **UP** → you benefit.

➡️ **LONG CALL**

Pattern 2

$$\boxed{\max(K - S, 0)}$$

Underlying goes **DOWN** → you benefit.

➡️ **LONG PUT**

Pattern 3

$$\boxed{S_T - S_0}$$

You gain when underlying rises and lose when it falls.

➡️ **LONG FUTURES**

Pattern 4

$$\boxed{-\max(K - S, 0)}$$

You lose when underlying falls below the strike.

➡️ **SHORT PUT**

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8. Apply it directly to this question

Baywhite's additional payment is:

$$\boxed{\max(\text{SIXV return} - 5\%, 0)}$$

Compare this with:

$$\boxed{\max(S - K, 0)}$$

They have the **same structure**.

Therefore:

$$\boxed{\text{LONG CALL}}$$

Answer:

B. The Structured Note has an embedded long call option contract with SIXV as the underlying.

⭐ One sentence to memorize

> **If the investor receives the upside above a specified level, but gets zero when the underlying is below that level → think LONG CALL.**

If you want, I can next explain **why a structured note can be decomposed into a bond + long call**, which is the next important CFA concept behind this question.

Which of the following statements best contrasts the credit risk of the Baywhite Financial LLC Structured Note with the counterparty credit risk of an investor © CFA Institute. For candidate use only. Not for distribution. 74 Learning Module 3 Derivative Benefits, Risks, and Issuer and Investor Uses entering into the embedded exchange-

traded derivative on a stand-alone basis? A. An investor in the Baywhite Structured Note assumes the credit risk of Baywhite Financial LLC for 20% of the note's face value, as the remaining 80% is principal protected. An investor entering into the SIXV derivative on a stand-alone basis

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Yes. This question is testing a **very important CFA distinction: credit risk of a structured note vs. counterparty credit risk of a derivative**.

Correct answer: C

> **An investor in the Baywhite Structured Note assumes the credit risk of Baywhite Financial LLC for 100% of the note's face value, while an investor entering into the SIXV derivative on a stand-alone basis assumes the counterparty credit risk of an exchange and its clearinghouse.**

Let's understand **why** very carefully.

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1. First: What is the Baywhite Structured Note?

Think of the note as:

$$\boxed{\text{Structured Note} = \text{Debt} + \text{Embedded Call Option}}$$

We already established that the embedded derivative is a **long call**.

But here's the important point:

Who owes the investor the money?

Baywhite Financial LLC.

The investor is lending money to Baywhite in exchange for the note.

So Baywhite is the **issuer/debtor**.

---

2. What does "80% principal protected" mean?

This is the trap in choices A and B.

The note says:

> **80% principal protected**

You might therefore think:

> "Okay, Baywhite's credit risk is only 20% because 80% is protected."

❌ **Wrong.**

"Principal protected" does **not** mean "free from issuer credit risk."

The protection itself is simply a **promise made by Baywhite**.

---

3. Imagine Baywhite goes bankrupt

Suppose you bought:

$$\$1,000$$

of face value.

The note says:

> "At maturity, I'll give you at least $800."

But who promised you that $800?

Baywhite.

Now suppose Baywhite becomes insolvent.

You can't say:

> "But my $800 is principal protected!"

The $800 protection is only as good as the **creditworthiness of Baywhite**.

Therefore, you have credit exposure to Baywhite for the **entire $1,000 face value**.

That's why:

$$\boxed{\text{Credit risk = 100% of face value}}$$

---

4. Why is A wrong?

Choice A says:

> Credit risk is only **20%** because 80% is principal protected.

This is incorrect.

The word **"protected"** does not mean "protected from issuer default."

It means:

> Under the terms of the contract, Baywhite promises to return 80% of face value at maturity, subject to Baywhite being able to meet its obligation.

So even the supposedly protected $800 is exposed to **Baywhite's default risk**.

Therefore:

❌ **A is wrong.**

---

5. Why is B wrong?

B says:

> Baywhite credit risk is only 80% because the remaining 20% is associated with the embedded derivative.

This is also incorrect.

Remember our decomposition:

$$\boxed{\text{Note} = \text{Debt} + \text{Derivative}}$$

You might think:

> "The derivative is separate, so I don't have Baywhite credit risk on that part."

But that's not how the investor's exposure works.

The investor owns **one note issued by Baywhite**.

Baywhite owes the investor whatever payment is due under the note.

Therefore, the investor has **issuer credit risk for the entire face value**.

❌ **B is wrong.**

---

6. Now the second half: stand-alone exchange-traded derivative

This is where **C** gets the second part right.

Suppose instead of buying Baywhite's structured note, you buy an **exchange-traded derivative** on SIXV.

Now you aren't simply relying directly on Baywhite.

Exchange-traded derivatives generally involve:

$$\boxed{\text{Exchange} + \text{Clearinghouse/Central Counterparty}}$$

The clearinghouse becomes the counterparty to the trades and helps manage counterparty credit risk through mechanisms such as margining and default procedures.

So instead of:

Investor → Baywhite

you have something more like:

Investor → Clearing system/CCP ← Other market participant

Therefore, the investor's counterparty credit risk is associated with the **central clearing structure**, rather than simply relying on the financial intermediary that issued a structured note.

---

7. The BIG difference

Let's put the two situations side-by-side.

Baywhite Structured Note

You buy the note:

$$\boxed{\text{Investor} \rightarrow \text{Baywhite}}$$

Baywhite promises to pay you according to the note terms.

Therefore:

$$\boxed{\text{Issuer credit risk} = \text{Baywhite}}$$

And importantly:

$$\boxed{\text{Credit exposure applies to 100\% of face value}}$$

Stand-alone exchange-traded derivative

You trade the derivative through an exchange/clearing system:

$$\boxed{\text{Investor} \rightarrow \text{Exchange/CCP}}$$

The clearinghouse stands between buyers and sellers and manages counterparty risk.

Therefore:

$$\boxed{\text{Counterparty credit risk} = \text{exchange/clearinghouse structure}}$$

---

8. Why the "100%" point is so important

This is probably the **main CFA exam trap**.

Don't think:

> 80% principal protected = 80% credit-risk-free.

Instead think:

> **Principal protection is a contractual payment promise from the issuer.**

If the issuer defaults, the protection may not be received.

Therefore:

$$\boxed{\text{100\% of face value is subject to issuer credit risk}}$$

---

9. Simple analogy

Imagine I give you a piece of paper saying:

> **"I promise to give you ₹80,000 after one year."**

You might call that **₹80,000 guaranteed**.

But suppose **I go bankrupt**.

Who was guaranteeing the ₹80,000?

I was.

So the ₹80,000 isn't magically protected from my bankruptcy.

That's exactly what happens with the structured note.

The **80% protection is a promise from Baywhite**.

🧠 CFA Exam Rule

Structured note

$$\boxed{\text{Investor bears issuer credit risk}}$$

Even if the note says:

- 80% principal protected
- 90% principal protected
- 100% principal protected

the investor still has **issuer credit risk**.

Exchange-traded derivative

$$\boxed{\text{Counterparty credit risk is reduced/managed through central clearing}}$$

Final answer

$$\boxed{\text{C}}$$

The critical distinction:

> **Principal protection ≠ protection from issuer default.**

So the Baywhite investor has **credit risk to Baywhite on 100% of the face value**, while the stand-alone exchange-traded derivative uses an **exchange/clearinghouse structure to manage counterparty credit risk**.

Which of the following statements best describes how an investor should evaluate the terms of the Baywhite Financial LLC Structured Note as compared with the stand-alone derivative price in order to make a recommendation? A. The Baywhite Financial LLC Structured Note issuance price of 2% above par value should be compared with the upfront premium for a six-month SIXV call option with an exercise price at 5% above the current SIXV spot price. B. The Baywhite Financial LLC Structured Note 20% Principal at Risk should be compared with the upfront premium for a six-month SIXV

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Correct Answer: A

This question is testing **what should actually be compared when evaluating a structured note against a stand-alone derivative**.

Let's go slowly.

---

1. What are we comparing?

The firm wants to compare:

Baywhite Structured Note

with

A stand-alone SIXV call option

We already know the embedded derivative is a:

$$\boxed{\text{Long SIXV Call}}$$

with an effective threshold/strike corresponding to a **5% increase** in SIXV.

The question is asking:

> **What feature of the structured note should we compare with the price (premium) of buying that call separately?**

---

2. Look at the three choices

A

> Structured Note issuance price of **2% above par** should be compared with the upfront premium for the SIXV call.

B

> **20% principal at risk** should be compared with the call premium.

C

> **2% above par + 20% principal at risk** should be compared with the call premium.

The key is understanding what **issuance price** means.

---

3. What does the investor actually pay?

The note has:

- Face value = $1,000
- Issuance price = **102% of face value**

Therefore:

$$\$1,000 \times 102\% = \boxed{\$1,020}$$

The investor pays **$1,020** for something with a face value of $1,000.

So the **extra cost** of buying the structured note is:

$$\$1,020 - \$1,000 = \boxed{\$20}$$

or:

$$\boxed{2\%}$$

This **2% premium over par** is the relevant upfront cost that can be compared with the premium of purchasing the stand-alone call.

---

4. Why don't we compare the 20% Principal at Risk?

This is the key trap.

The **20% principal at risk** is not a price.

It tells you about the **downside protection of the note**.

Remember:

$$\boxed{80\% \text{ protected}}$$

and

$$\boxed{20\% \text{ at risk}}$$

That is a **risk/payoff characteristic**, not the amount the investor pays for the embedded derivative.

The call option's premium, on the other hand, is an **upfront price**.

So we need to compare:

$$\boxed{\text{Cost of structured note}}$$

with

$$\boxed{\text{Cost of stand-alone call}}$$

Not:

$$\text{Risk percentage vs. option premium}$$

---

5. Why C is wrong

C says we should compare:

$$2\% + 20\%$$

against the call premium.

But that would be mixing two completely different concepts.

2% above par

= pricing/cost

20% principal at risk

= risk/payoff characteristic

You don't add them together to determine the cost of the embedded call.

Therefore:

❌ C is wrong.

---

6. Think about it like buying a phone 📱

Imagine:

- Phone's normal value = ₹100,000
- You pay ₹102,000

The extra ₹2,000 is the **premium you're paying**.

Now suppose the phone has a 20% chance of losing some value.

Would you say:

> "The price premium is ₹2,000 + 20%?"

No.

The 20% describes **risk**, while ₹2,000 describes **price**.

Same concept here.

---

7. The CFA way to think about it

The structured note is essentially:

$$\boxed{\text{Debt component} + \text{Embedded call}}$$

To determine whether the structured note is attractively priced relative to buying the derivative separately, you compare the **additional amount paid for the note** with the **premium for the equivalent stand-alone derivative**.

The note costs:

$$102\% - 100\% = \boxed{2\% \text{ above par}}$$

The equivalent stand-alone derivative costs:

$$\boxed{\text{Six-month SIXV call premium}}$$

Therefore:

$$\boxed{\text{2\% issuance premium vs. call premium}}$$

---

8. One very important distinction

Don't confuse these three numbers:

Item	Meaning
102% issuance price	What investor pays
2% above par	Extra amount paid over face value
20% principal at risk	Downside exposure of the note

The question asks you to compare **price with price**.

Therefore:

$$\boxed{2\%\ \text{premium} \quad \text{vs.} \quad \text{call option premium}}$$

✅ Final Answer: A

CFA exam shortcut:

> **When comparing a structured note with the equivalent stand-alone derivative, compare the note's pricing premium/cost with the derivative's premium—not the note's principal-at-risk percentage.**

So:

$$\boxed{\text{A}}$$

2% above par → compare with the upfront premium of the six-month SIXV call with a 5%-above-spot exercise level.

Can you help me understand what is stand-alone derivative strategie

Absolutely. This is a **very important idea** for understanding the Baywhite question.

Let's start from **zero**.

1. What does "stand-alone derivative" mean?

A **stand-alone derivative** is simply a derivative that you **buy or sell separately**, rather than having it embedded inside another financial product.

For example:

Stand-alone call option

You go to the derivatives market and say:

> "I want to buy a 6-month call option on SIXV."

You pay an option premium, and that's your investment in the derivative.

There is **no structured note attached to it**.

That's a **stand-alone derivative**.

---

2. Compare it with the Baywhite structured note

The Baywhite product is different.

You aren't directly buying the call option.

Instead, you're buying:

$$\boxed{\text{Structured Note}}$$

And inside that note there is an **embedded call option**.

Think of it like this:

Baywhite

You → Buy Structured Note

Inside the note:

$$\boxed{\text{Debt component} + \text{Call option}}$$

The call is **embedded** inside the note.

Stand-alone derivative

You simply do:

You → Buy Call Option

There is no debt/security wrapped around it.

That's why we call it:

$$\boxed{\text{Stand-alone derivative}}$$

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3. Why does CFA compare the two?

Because an investor might ask:

> "Why should I buy this complicated structured note instead of simply buying the derivative myself?"

That's exactly what the case is asking you to evaluate.

For example:

Option 1 — Buy Baywhite note

You pay:

$$102\% \text{ of face value}$$

You get:

- 80% principal protection
- Exposure to SIXV above 5%
- Credit exposure to Baywhite

Option 2 — Buy the derivative separately

You could instead buy a:

$$\boxed{\text{6-month SIXV call}}$$

You pay an **option premium**.

You then separately decide what to do with your remaining money—for example, invest it in another asset.

---

4. What does "stand-alone derivative strategy" mean?

This is slightly broader than just "stand-alone derivative."

A **stand-alone derivative strategy** means using derivatives **by themselves or in a combination**, rather than having them embedded in a structured product.

For example:

Strategy 1 — Long call

You buy a call on SIXV.

You expect SIXV to rise.



Strategy 2 — Long put

You buy a put on SIXV.

You expect SIXV to fall or want downside protection.



Strategy 3 — Futures hedge

Suppose you own an asset and want to protect against a decline.

You could sell futures.



That's a stand-alone derivative strategy.

---

5. Why is the word "stand-alone" important?

Because there are two different ways a derivative can appear.

A. Embedded derivative

Derivative is **inside another financial instrument**.

Example:



containing a:



B. Stand-alone derivative

You trade the derivative **directly**.

Example:



No structured note.

---

6. Let's use a simple real-life analogy

Imagine you want a camera.

Option A — Camera inside a phone

You buy a smartphone.

The phone contains a camera.

You didn't buy the camera separately.

That's like an **embedded derivative**.

Option B — Buy the camera separately

You buy a standalone camera.

That's like a **stand-alone derivative**.

The camera exists independently.

---

7. Now apply this directly to Baywhite

Baywhite says:

> "I'll give you 80% principal protection plus an additional payment based on SIXV's return above 5%."

We identified:

$$\boxed{\text{Additional payment} = \text{Long Call}}$$

So:

Baywhite product

$$\boxed{\text{Structured Note}}$$

contains:

$$\boxed{\text{Long SIXV Call}}$$

Therefore:

Long call = embedded derivative.

But an investor could instead go to the derivatives market and buy:

$$\boxed{\text{6-month SIXV Call}}$$

directly.

That is the:

$$\boxed{\text{Stand-alone derivative}}$$

---

8. This explains the previous question

The previous question asked:

> How should we compare the Baywhite note with the stand-alone derivative?

The idea is:

Baywhite gives you a structured product containing a call.

versus

You could buy the equivalent call separately.

So we compare the economics.

Baywhite:

\text{2% premium above par}

versus

Stand-alone:

$$\text{Call option premium}$$

That's why the answer was **A**.

⭐ The CFA concept you should remember

Whenever you see:

Structured product

Think:

$$\boxed{\text{Traditional security} + \text{Embedded derivative}}$$

When you see:

Stand-alone derivative

Think:

$$\boxed{\text{Derivative traded separately}}$$

And in this case:

$$\boxed{\text{Baywhite Note} = \text{Debt} + \text{Embedded Long SIXV Call}}$$

while:

$$\boxed{\text{Alternative} = \text{Buy SIXV Call Directly}}$$

One-line memory trick:

> **Embedded = derivative hidden inside another product. Stand-alone = derivative bought/sold directly.**

That distinction will make the next questions in this CFA reading **much easier**.

Match the following definitions with their corresponding forward pricing or
valuation component.
1. Equal to the difference between the current spot price (adjusted by remaining costs and benefits through maturity) and the present value of the forward price
A. Forward contract MTM value at inception, V0(T)
2. Future value of the underlying asset spot price (S0) compounded at the

Show more

Yes. Let's solve this **step by step**, because the wording is deliberately complicated.

We need to match the **three definitions** with:

- **A.** Forward contract MTM value at inception, $V_0(T)$
- **B.** Forward contract MTM value after inception, $V_t(T)$
- **C.** Forward price, $F_0(T)$

Step 1: Understand the three terms first

1. Forward price $F_0(T)$

This is the **price agreed today for the asset to be delivered in the future**.

Think:

> "What delivery price should we agree on today so that neither side has an advantage?"

That's the **forward price**.

---

2. Forward contract value at inception $V_0(T)$

When a new forward contract is entered into at the **fair forward price**, its value is:

$$\boxed{V_0(T) = 0}$$

Why?

Because neither the buyer nor seller should have an advantage at the beginning.

---

3. Forward contract value later $V_t(T)$

After some time has passed, market conditions may change.

The forward contract may now be worth:

- positive to the buyer, or
- negative to the buyer.

That's its **mark-to-market (MTM) value**.

Now match the definitions.

Definition 1

> "Equal to the difference between the current spot price (adjusted by remaining costs and benefits through maturity) and the present value of the forward price."

This is describing the **current value of an existing forward contract**.

It is comparing:

- value of the underlying today, adjusted for costs/benefits
- present value of the contracted forward price

That's a **mark-to-market value**.

And because it says **current** rather than inception:

$$\boxed{1 \rightarrow \mathrm{B}}$$

Answer:

1 → B. Forward contract MTM value between inception and maturity, $V_t(T)$

Definition 2

> "Future value of the underlying asset spot price (S_0) compounded at the risk-free rate incorporating the present value of the costs and benefits of asset ownership."

This is describing the **forward price**.

The basic forward pricing relationship is:

$$\boxed{F_0(T) = S_0(1+r)^T}$$

when there are no costs or benefits.

With costs and benefits, conceptually:

$$\boxed{\text{Forward price} = \text{Future value of spot} + \text{costs} - \text{benefits}}$$

So:

$$\boxed{2 \rightarrow \text{C}}$$

Answer:

2 → C. Forward price, $F_0(T)$

Definition 3

> "Under no-arbitrage conditions for a given underlying spot price, S_0, adjusted by the costs and benefits, risk-free rate r, and forward price $F_0(T)$, this should be equal to zero."

What should equal zero when we enter a **fair forward contract**?

The answer is:

$$\boxed{V_0(T) = 0}$$

The forward contract's value at inception is zero under no-arbitrage conditions.

Therefore:

$$\boxed{3 \rightarrow \text{A}}$$

Answer:

3 → A. Forward contract MTM value at inception, $V_0(T)$

✅ Final Matching

Definition	Answer	Why
1	B	Current MTM value of an existing forward
2	C	Determines the forward price
3	A	Fair forward has zero value at inception

Therefore:

$$\boxed{1-B,\quad 2-C,\quad 3-A}$$

🧠 The easiest way to remember this

Think of the timeline:

$$\boxed{\text{Today} \quad \longrightarrow \quad \text{Later} \quad \longrightarrow \quad T}$$

TODAY — Inception

A fair forward has:

$$\boxed{V_0 = 0}$$

➡️ **A**

TODAY — Determine the agreed delivery price

We calculate:

$$\boxed{F_0(T)}$$

➡️ **C**

LATER — Contract already exists

Market conditions have changed, so we calculate:

$$\boxed{V_t(T)}$$

➡️ **B**

⭐ CFA shortcut

When you see:

"equal to zero under no-arbitrage at inception"

$\rightarrow V_0(T)$

When you see:

"future value of spot adjusted for costs and benefits"

$\rightarrow F_0(T)$

When you see:

"current value / MTM of an existing forward"

$\rightarrow V_t(T)$

So memorize:

$$\boxed{V_0 = 0, \qquad F_0 = \text{fair delivery price}, \qquad V_t = \text{current MTM value}}$$

Can you help me with example for each match ?

Absolutely. The easiest way to understand these three is to use **one simple forward contract** and watch what happens at **three different points in time**.

Let's use a stock with **no dividends or other costs** first.

Our example

Suppose today:

- Current stock price $S_0 = \$100$
- Risk-free rate = **5% per year**
- Maturity = **1 year**

We enter a 1-year forward contract.

The fair forward price is:

$$F_0(T) = S_0(1+r)^T$$

$$= 100(1.05)^1$$

$$= \boxed{\$105}$$

So the forward contract says:

> **One year from now, I will buy the stock for $105.**

Now let's see how this relates to each match.

1️⃣ Match 2 → C: Forward Price $F_0(T)$

Remember definition 2:

> Future value of the underlying spot price, compounded at the risk-free rate, incorporating costs and benefits.

We calculated:

$$F_0(T) = 100(1.05) = \$105$$

So:

$$\boxed{F_0(T) = \$105}$$

What does $105 mean?

It is **not** the current value of the contract.

It is the **agreed delivery price** that makes the contract fair today.

Think:

> "If I agree today to buy this stock one year from now, what price should I agree to?"

Answer:

$$\boxed{\$105}$$

Therefore:

$$\boxed{\text{Definition 2} \rightarrow \text{C}}$$

2️⃣ Match 3 → A: Forward value at inception $V_0(T)$

Now we actually enter the forward contract.

The agreed forward price is:

$$F_0(T) = \$105$$

Immediately after entering the contract, what is the contract worth?

Answer:

$$\boxed{V_0(T) = 0}$$

Why?

Because the forward price was chosen to be the **fair price**.

Neither party is getting an advantage.

Think about it this way

You agree:

> "I will buy this stock in one year for $105."

The seller says:

> "Okay."

At that exact moment, neither of you has made a profit.

Therefore:

$$\boxed{V_0 = 0}$$

That's why:

$$\boxed{\text{Definition } 3 \rightarrow \text{A}}$$

3️⃣ Match 1 → B: Forward MTM value after inception

Now let's travel **six months into the future**.

Suppose the stock price has increased dramatically.

The stock is now worth:

$$\$120$$

But you have a forward contract that allows you to buy it for:

$$\$105$$

That's fantastic for you!

You can buy something worth $120 for only $105.

Your economic benefit is approximately:

$$120 - 105 = \boxed{\$15}$$

So your forward contract now has a **positive value**.

This is called the:

$$\boxed{\text{Mark-to-market (MTM) value}}$$

or:

$$\boxed{V_t(T)}$$

Technically, because the $105 payment occurs at maturity, we discount the difference to today's value.

But for understanding the concept, think:

$$\boxed{\text{Current value} \approx \$15}$$

Therefore:

$$\boxed{\text{Definition } 1 \to \text{B}}$$

Put the whole story together

Here's the timeline:

```
TODAY                          6 MONTHS                  1 YEAR
  |                               |                        |
  |                               |                        |
```



Now match each definition

Definition 1

> Difference between current spot value and PV of forward price.

We're **after inception**.

Therefore:

$$\boxed{1 \to B}$$

MTM value between inception and maturity.

Definition 2

> Future value of spot price adjusted for interest, costs and benefits.

We're calculating the **fair delivery price**.

Therefore:

$$\boxed{2 \to C}$$

Forward price.

Definition 3

> Under no-arbitrage conditions, this should equal zero.

That's the value **when the contract is first entered**.

Therefore:

$$\boxed{3 \to A}$$

Forward value at inception.

⭐ One very simple way to remember

Think about three questions:

Question 1: "What price should we agree on?"

That's:

$$\boxed{F_0(T)}$$

Forward price → C

Question 2: "What is the contract worth when we first sign it?"

$$\boxed{V_0 = 0}$$

Value at inception → A

Question 3: "What is our existing contract worth now?"

$$\boxed{V_t}$$

MTM value → B

Final memory table

What are you asking?	Formula/concept	Match
What should the future delivery price be?	$F_0(T)$	**C**
What is the contract worth when we enter it?	$V_0 = 0$	**A**
What is the contract worth later?	V_t	**B**

So the answer remains:

$$\boxed{\textbf{1-B,} \quad \textbf{2-C,} \quad \textbf{3-A}}$$

The most important distinction

Forward price ≠ forward value.

- **Forward price** F_0 = the **price written into the contract**
- **Forward value** V_0 = what the contract is **worth**
- **Forward value** V_t = what the contract is **worth later**

That distinction is extremely important for CFA Level I Derivatives.

A Baywhite client has entered into a long six-month MXN/USD FX forward contract—that is, an agreement to sell MXN and buy USD. The MXN/USD spot exchange rate at inception is 19.8248 (MXN19.8248 = USD1), the six-month MXN risk-free rate is 4.25%, and the six-month USD risk-free rate is 0.5%. Baywhite's market strategist predicts that the Mexican central bank (Banco de Mexico) will surprise the market with a 50 bp short-term rate cut at its upcoming meeting. Which of the following statements best describes how the client's existing FX forward contract will be impacted if this prediction is realized and

Show more

Yes. This one is testing **two separate ideas at the same time**:

1. How an interest-rate change affects the **FX forward price**
2. How that change affects the **MTM value of an existing long FX forward**

The key is not to mix those two.

Correct answer: B

Let's build it from scratch.

---

1. First understand the position

The client has a **long six-month MXN/USD forward**.

The question tells us exactly what that means:

> Agreement to **sell MXN and buy USD**.

So the client is:

$$\boxed{\text{Long USD / Short MXN}}$$

This is very important.

The client wants the **USD to become more valuable relative to MXN**.

---

2. What does MXN/USD = 19.8248 mean?

The quote is:

$$\boxed{19.8248 \text{ MXN} = 1 \text{ USD}}$$

So:

> One USD costs 19.8248 MXN.

If the exchange rate goes from:

$$19.8248 \rightarrow 20.50$$

then USD has become **more expensive** in terms of MXN.

That means:

$$\boxed{\text{USD strengthened / MXN weakened}}$$

And that's good for our client because the client is **long USD**.

---

3. Now look at the interest rates

Initially:

Currency	6-month risk-free rate
MXN	4.25%
USD	0.50%

The interest-rate differential is:

$$4.25\% - 0.50\% = \boxed{3.75\%}$$

Now suppose Banco de Mexico cuts MXN rates by **50 bp**.

50 bp = 0.50%.

So MXN rate becomes:

$$4.25\% - 0.50\% = \boxed{3.75\%}$$

Now the differential is:

$$3.75\% - 0.50\% = \boxed{3.25\%}$$

So the MXN-USD interest-rate differential **falls**.

---

4. What happens to the FX forward rate?

For an FX forward quoted as:

$$\boxed{\text{MXN/USD}}$$

the approximate forward pricing relationship is:

$$\boxed{F_0 = S_0 \times \frac{1 + r_{MXN}}{1 + r_{USD}}}$$

So if the MXN interest rate falls, the numerator becomes smaller.

Therefore:

$$\boxed{F_0 \text{ decreases}}$$

So **A is correct about the direction of the forward rate**.

But A asks about the **contract forward rate being adjusted downward**.

That's not the best answer for the client's **existing contract**.

Why?

Because the client already locked in a forward price at inception.

The contract's original forward price doesn't get rewritten.

Instead, the **market value of the existing contract changes**.

That's the key distinction.

---

5. What happens to the EXISTING contract?

This is where **B** comes in.

The client is:

$$\boxed{\text{Long USD}}$$

Suppose the original forward rate was higher than the new fair forward rate.

After the MXN rate cut, the new fair forward rate goes **down**.

Imagine:

Original forward contract

Client agreed:

$$\boxed{20.10 \text{ MXN/USD}}$$

But after the rate cut, a new six-month forward might be:

$$\boxed{19.95 \text{ MXN/USD}}$$

The client's existing contract lets them buy USD at:

$$20.10$$

But a new investor can buy USD forward at:

$$19.95$$

That's bad for our client.

So wouldn't that mean a loss?

Careful!

We need to check the direction of the contract.

The client is **selling MXN and buying USD**.

If the agreed rate is quoted as MXN per USD, the client is effectively paying MXN for USD.

A higher MXN/USD forward rate means the client has to give up **more MXN for each USD**.

So a lower forward rate is actually **better** for someone wanting to buy USD.

Therefore:

$$\boxed{\text{New forward rate falls} \Rightarrow \text{existing long USD position gains}}$$

Thus:

$$\boxed{\text{MTM gain}}$$

---

6. Let's make it extremely simple with an example

Suppose you have a contract allowing you to buy:

$$\$1$$

for:

$$20 \text{ MXN}$$

Now imagine the market changes and USD can only be bought forward for:

$$19 \text{ MXN}$$

Would your old contract be valuable?

No — wait, let's be precise.

Your contract says:

> **You pay 20 MXN to receive 1 USD.**

But the market says:

> **You can pay only 19 MXN to receive 1 USD.**

So your contract is **worse than the market**.

Therefore, the value of your old contract is negative.

So this would be a **loss**.

That suggests we need to revisit the direction carefully.

7. Here's the crucial CFA point

For an FX forward quoted:

$$\boxed{\text{MXN/USD}}$$

and a client who is:

$$\boxed{\text{selling MXN and buying USD}}$$

the forward contract payoff depends on the difference between the **spot/forward value of USD** and the contracted rate.

If the fair forward rate decreases, the right to buy USD at the **old, higher rate** becomes less valuable.

Therefore:

$$\boxed{\text{Existing contract loses value}}$$

So the correct answer is actually:

✅ C

I want to correct the earlier direction clearly: **C is the best answer.**

---

8. Why C is correct

The MXN interest rate falls:

$$4.25\% \rightarrow 3.75\%$$

Therefore the MXN/USD forward rate falls.

The client is locked into the **old forward rate**.

The new market forward rate is lower.

Because the client has agreed to buy USD at the old, higher MXN/USD rate, the existing contract becomes less attractive.

Therefore:

$$\boxed{\text{MTM loss}}$$

So:

$$\boxed{\mathbf{C}}$$

---

9. Why A is not the best answer

A says:

> The contract forward rate will be adjusted downward.

The **market forward rate** will indeed be lower.

But the client's **existing contractual forward rate does not change**.

That's the key.

The original contract is fixed.

What changes is:

$$\boxed{\text{MTM value}}$$

So A confuses:

new market forward price

with

existing contract's forward price.

---

10. Why B is wrong

B says:

> Client realizes an MTM gain.

But the client is locked into buying USD at the old rate.

After the MXN rate falls, the new fair MXN/USD forward rate falls.

Therefore, the client's existing contract becomes less valuable.

$$\boxed{\text{MTM loss}}$$

So B is incorrect.

⭐ CFA exam framework

Whenever you see an FX forward question, do these **three steps**:

Step 1 — Identify the quote

$$\boxed{\text{MXN/USD} = \text{MXN per USD}}$$

Step 2 — Identify the position

Client:

$$\boxed{\text{Sell MXN / Buy USD}}$$

Therefore:

$$\boxed{\text{Long USD}}$$

Step 3 — Determine what happens to the NEW forward rate

$$F = S \times \frac{1 + r_{MXN}}{1 + r_{USD}}$$

MXN rate ↓

↓

MXN/USD forward rate ↓

↓

Existing contract has **higher contracted rate**

↓

Client is buying USD at a relatively expensive rate

↓

$$\boxed{\text{MTM LOSS}}$$

Final answer

$$\boxed{\mathbf{C}}$$

The key distinction to memorize:

> **The market forward rate changes, but the forward rate written into an existing contract does not change. The change in the market forward rate causes the existing contract's MTM value to change.**

So here:

$$\boxed{\text{MXN rate} \downarrow \rightarrow \text{MXN/USD forward rate} \downarrow \rightarrow \text{existing long USD forward} \rightarrow \text{M}}$$

Can you help me with numerical example ?

Absolutely. This question becomes much easier if we use actual numbers. The **most important thing** is to see what happens to the **existing contract** when the interest rate changes.

Let's use a simplified numerical example.

---

1. Start with the original situation

Suppose:

$$S_0 = 19.8248 \text{ MXN/USD}$$

MXN interest rate:

$$r_{MXN} = 4.25\%$$

USD interest rate:

$$r_{USD} = 0.50\%$$

Maturity:

$$T = 0.5 \text{ years}$$

The approximate FX forward pricing formula is:

$$F_0 = S_0 \frac{1 + r_{MXN} T}{1 + r_{USD} T}$$

So:

$$F_0 = 19.8248 \times \frac{1 + (0.0425)(0.5)}{1 + (0.005)(0.5)}$$

$$F_0 = 19.8248 \times \frac{1.02125}{1.0025}$$

$$\boxed{F_0 \approx 20.1955 \text{ MXN/USD}}$$

So the client enters into a contract to:

> **Sell MXN and buy USD at 20.1955 MXN per USD.**

---

2. Now Banco de Mexico cuts rates

The central bank unexpectedly cuts the MXN rate by:

$$50 \text{ bp} = 0.50\%$$

So:

$$4.25\% - 0.50\% = \boxed{3.75\%}$$

USD rate remains:

$$0.50\%$$

Now calculate the **new market forward rate**.

$$F_{new} = 19.8248 \times \frac{1 + (0.0375)(0.5)}{1 + (0.005)(0.5)}$$

$$F_{new} = 19.8248 \times \frac{1.01875}{1.0025}$$

$$\boxed{F_{new} \approx 20.1465}$$

So:

Before rate cut:

$$\boxed{20.1955}$$

After rate cut:

$$\boxed{20.1465}$$

Therefore:

$$\boxed{\text{Forward rate decreased}}$$

---

3. But here's the REALLY important part

The client's existing contract still says:

$$\boxed{20.1955}$$

It does **not** change to 20.1465.

The new market forward is:

$$\boxed{20.1465}$$

So imagine two contracts now exist:

Client's old contract

> Buy USD at **20.1955 MXN/USD**

New market contract

> Buy USD at **20.1465 MXN/USD**

Which one would you rather have?

Obviously:

$$\boxed{20.1465}$$

because you need fewer pesos to buy each dollar.

Therefore, the client's old contract is now **less valuable**.

That's why the client has an:

$$\boxed{\text{MTM LOSS}}$$

---

4. Let's make the difference even more obvious

Suppose the client needs to buy:

$$\$1,000,000$$

under the forward contract.

Under the client's old contract:

$$1,000,000 \times 20.1955$$

$$= \boxed{20,195,500 \text{ MXN}}$$

The client must pay:

MXN 20.1955 million

Under today's new market forward:

$$1,000,000 \times 20.1465$$

$$= \boxed{20,146,500 \text{ MXN}}$$

A new market participant can buy the same USD 1 million for:

MXN 20.1465 million

Difference:

$$20,195,500 - 20,146,500$$

$$= \boxed{49,000 \text{ MXN}}$$

The client's existing contract is approximately **MXN 49,000 worse** than the new market forward, before discounting/precise valuation conventions.

Therefore:

$$\boxed{\text{MTM Loss}}$$

---

5. Why did the interest-rate cut cause this?

This is the beautiful part.

Initially:

$$r_{MXN} = 4.25\%$$

After the cut:

$$r_{MXN} = 3.75\%$$

Therefore the MXN interest-rate advantage over USD became smaller.

That caused the:

$$\boxed{\text{MXN/USD forward rate to fall}}$$

And because our client had agreed to buy USD at the **old higher rate**, the existing contract became less valuable.

---

6. Think of it like buying something

Imagine you agreed today:

> "Six months from now, I'll buy a laptop for ₹100,000."

Six months later, the same laptop can be bought for:

$$₹95,000$$

Your contract still says:

$$₹100,000$$

Would your contract be valuable?

No.

You are stuck paying:

$$₹100,000$$

when the market price is:

$$₹95,000$$

So your contract has a:

$$\boxed{₹5{,}000 \text{ loss}}$$

Exactly the same idea is happening with the FX forward.

---

7. Now connect it directly to the three answers

A

> Forward rate will be adjusted downward.

The **new market forward rate** does decrease.

But the client's existing contractual forward rate doesn't get "adjusted."

❌ **A is not the best answer.**

B

> Client realizes an MTM gain.

No.

The client is locked into:

$$20.1955$$

while the market is now:

$$20.1465$$

The client is paying more MXN per USD than necessary.

❌ **B is wrong.**

C

> Lower interest-rate differential causes the client to realize an MTM loss.

Exactly.

$$\boxed{r_{MXN} \downarrow \Rightarrow F_{MXN/USD} \downarrow \Rightarrow \text{existing long USD forward loses value}}$$

✅ **C is correct.**

⭐ One thing I want you to remember

There are **two different forward rates** in this question:

1. Contracted forward rate

The rate the client locked in:

$$\boxed{20.1955}$$

This does NOT change.

2. Current market forward rate

The rate available after the interest-rate change:

$$\boxed{20.1465}$$

This DOES change.

The difference between these two determines the **MTM value** of the existing contract.

CFA exam shortcut

Whenever you see:

> **"Existing forward contract"**

don't change the original contractual forward price.

Instead ask:

> **"What is the new market forward price, and is the existing contract better or worse than the new market price?"**

Here:

$$\boxed{20.1955 > 20.1465}$$

Client has to buy USD at the **higher** rate.

Therefore:

$$\boxed{\textbf{MTM LOSS} \rightarrow \textbf{C}}$$

4. A client seeking advice on her fixed-income portfolio observes the price and
yield-to-maturity of one-year (r1) and two-year (r2) annual coupon government
benchmark bonds currently available in the market. Which of the following statements best describes how the analyst can determine a breakeven reinvestment
rate in one year's time to help decide whether to invest now for one or two years?
A. As the two-year rate involves intermediate cash flows, divide the square root

Show more

Yes. This is a **Fixed Income / spot-rate / forward-rate** question, and the tricky part is understanding **why we need the 2-year zero/spot rate rather than the 2-year coupon bond YTM**.

Correct answer: C

Let's build it from the beginning.

---

1. What is the question trying to find?

The client is deciding:

Option 1: Invest for 1 year

Buy a 1-year government bond.

After one year, the money matures, and then the client can reinvest it for another year.

So the total two-year growth is:

$$\boxed{(1+r_1)(1+\text{reinvestment rate})}$$

Option 2: Invest for 2 years

Buy a 2-year government bond and hold it for two years.

But the problem is:

> The 2-year bond is a **coupon bond**, so it has an intermediate coupon payment after one year.

Therefore, its YTM r_2 is **not directly the same thing as the 2-year spot/zero rate**.

That's the key issue.

---

2. What does "breakeven reinvestment rate" mean?

It means:

> **What rate would I need to earn in the second year after investing in the 1-year bond so that I end up exactly as well off as if I had invested in the 2-year bond?**

So we're looking for the one-year rate **one year from now** that makes the two strategies equivalent.

This is essentially a **1-year forward rate starting one year from now**.

---

3. Why can't we simply use r_2?

This is the biggest trap.

The 2-year bond has **coupon payments**.

For example, suppose:

- Face value = $1,000
- Coupon = 5%
- Maturity = 2 years

The cash flows are:

Time	Cash flow
Year 1	$50
Year 2	$1,050

There is a cash flow at **Year 1**.

Therefore, the 2-year YTM is a rate that discounts **both** cash flows:

$$P = \frac{50}{(1+r_2)} + \frac{1,050}{(1+r_2)^2}$$

So r_2 is a **single rate that makes the PV of multiple cash flows equal to the bond price**.

It is not necessarily the same as the true 2-year zero-coupon rate.

---

4. What do we actually need?

We need a rate that represents the return from:

$$\boxed{\text{Today} \rightarrow \text{Year 2}}$$

for a **single cash flow at Year 2**.

That's a **zero-coupon/spot rate**.

Call it:

$$\boxed{z_2}$$

So first we need to determine z_2.

That's exactly why **C** says:

> "first substitute the one-year rate r_1 into the two-year bond price equation to solve for the two-year spot or zero rate z_2"

---

5. Let's use a numerical example

Suppose the market gives us:

1-year bond

$$r_1 = 4\%$$

2-year coupon bond

Suppose:

- Face value = $1,000
- Coupon = 5%
- Price = $1,018.87

So the 2-year bond has cash flows:

$$\$50 \text{ at Year } 1$$

and

$$\$1,050 \text{ at Year } 2$$

---

6. First find z_2

The bond price must equal the PV of its cash flows.

We know the one-year spot rate:

$$z_1 = r_1 = 4\%$$

Therefore:

$$1018.87 = \frac{50}{1.04} + \frac{1050}{(1+z_2)^2}$$

First calculate the first cash flow:

$$\frac{50}{1.04} = 48.0769$$

Therefore:

$$1018.87 - 48.0769 = 970.7931$$

So:

$$970.7931 = \frac{1050}{(1+z_2)^2}$$

Therefore:

$$(1+z_2)^2 = \frac{1050}{970.7931}$$

$$1+z_2 = \sqrt{\frac{1050}{970.7931}}$$

So approximately:

$$\boxed{z_2 \approx 4.0\%}$$

In this particular numerical example, the rates happen to line up conveniently.

7. Now find the breakeven reinvestment rate

We know:

$$z_1 = r_1 = 4\%$$

and:

$$z_2 = 4\%$$

The relationship between spot rates and the one-year forward rate is:

$$\boxed{(1 + z_1)(1 + f_{1,2}) = (1 + z_2)^2}$$

Here:

$$(1 + 4\%)(1 + f) = (1 + 4\%)^2$$

Therefore:

$$1 + f = 1.04$$

$$\boxed{f = 4\%}$$

So if you invest for one year at 4%, then reinvest for another year at 4%, your two-year compounded return equals the two-year zero-coupon return.

---

8. Why C is exactly right

C says:

> First use r_1 in the 2-year coupon bond pricing equation to solve for z_2.

Correct.

Then:

$$\boxed{(1 + r_1)(1 + \text{breakeven reinvestment rate}) = (1 + z_2)^2}$$

Correct.

Then solve for the reinvestment rate.

Therefore:

$\boxed{\mathbf{C}}$

---

9. Why A is wrong

A says:

> Divide the square root of $(1 + r_2)$ by $(1 + r_1)$...

The problem is that r_2 **is the YTM of a coupon bond**.

It is **not necessarily the 2-year spot rate**.

You cannot directly use r_2 in the forward-rate equation.

You need:

$$\boxed{z_2}$$

not:

$$r_2$$

Therefore:

❌ **A is wrong.**

---

10. Why B is wrong

B says:

$$(1 + r_1)(1 + \text{breakeven rate}) = (1 + r_2)^2$$

Again, the problem is r_2.

The 2-year bond has an intermediate coupon payment.

Therefore its YTM isn't necessarily the appropriate rate for a two-year zero-coupon cash flow.

You need to first bootstrap the **2-year spot rate**.

So:

❌ **B is wrong.**

⭐ The key CFA concept

This question is really testing:

$$\boxed{\text{Coupon bond YTM} \neq \text{Spot rate}}$$

in general.

For a coupon bond:

$$\boxed{r_2 = \text{YTM}}$$

but we need:

$$\boxed{z_2 = \text{2-year spot rate}}$$

because the breakeven reinvestment calculation deals with the **single Year-2 cash flow**.

🧠 Memorize this process

When you see:

> **1-year YTM + 2-year coupon bond → find breakeven reinvestment rate**

Think:

Step 1

Find:

$$\boxed{z_1 = r_1}$$

Step 2

Use the 2-year bond price equation to **bootstrap** z_2.

Step 3

Calculate the one-year forward rate:

$$\boxed{(1 + z_1)(1 + f_{1,2}) = (1 + z_2)^2}$$

Step 4

Solve for:

$$\boxed{f_{1,2}}$$

Therefore:

$$\boxed{\text{C}}$$

One-line memory trick:

> **If the 2-year instrument is a coupon bond, don't use its YTM directly to calculate the forward/reinvestment rate—first get the 2-year spot rate.**

Can you get me the precise document for this

Yes. I found the **precise source containing this exact question**.

It is from the **CFA Institute Level I 2024 Derivatives & Alternative Investments** material, **Learning Module 5: Pricing and Valuation of Forward Contracts and for an Underlying with Varying Maturities**. The exact question appears on **page 126** of the material. Zrive +1

The exact question you gave is reproduced in the source, including choices **A, B, and C**. Zri...

The precise section you want

CFA Level I 2024 — Derivatives & Alternative Investments

Learning Module 5:
Pricing and Valuation of Forward Contracts and for an Underlying with Varying Maturities

The relevant concept is **Implied Forward Rate (IFR) / breakeven reinvestment rate**. The curriculum explains that the breakeven reinvestment rate linking a short-dated and long-dated **zero-coupon** bond is an implied forward rate. studylib....

CFA Level I 2024 Derivatives & Alternative Investments PDF — relevant source

One important correction to my previous explanation

The **official CFA explanation** specifically says that because the breakeven reinvestment involves a **single cash flow at maturity**, you first use the one-year rate r_1 in the two-year **coupon bond price equation** to solve for the two-year spot/zero rate z_2. Then:

$$(1 + r_1)(1 + \text{breakeven reinvestment rate}) = (1 + z_2)^2$$

That is why **C is correct**. studylib.…

If by **“precise document”** you mean **the exact CFA PDF/book from which all these Baywhite questions are taken**, the source I found is the **2024 Level I Derivatives & Alternative Investments volume**.

No i mean can you create a nice document of the explanation so that it would be very helpfull.

Yes — I misunderstood you earlier. You wanted a **study document containing the explanations and numerical examples**, not the source document.

I’ve created a structured CFA Level I study guide covering:

- Forward price vs. forward value vs. MTM value
- Numerical examples for each
- Stand-alone vs. embedded derivatives
- Baywhite structured note and long-call logic
- Structured-note issuer credit risk
- Structured note vs. stand-alone derivative pricing
- FX forward numerical example and MTM gain/loss
- Breakeven reinvestment rate / implied forward rate
- Why the correct answer is **C** in the fixed-income question
- CFA exam shortcuts and a final decision table

The forward-rate concepts are also consistent with the current CFA Institute refresher material. CFA Instit… +1

Download the CFA Level I Derivatives Study Guide

CFA_Level_I_Derivatives_Forward_Pricing_Study_Guid…
PDF

Baywhite Financial seeks to gain a competitive advantage by making margin loans at fixed rates for up to 60 days to its investor clients. Since Baywhite borrows at a variable one-month market reference rate to finance these client

loans, the firm enters into one-month FRA contracts on one-month MRR to hedge the interest rate exposure of its margin loan book. Which of the following statements best describes Baywhite's interest rate exposure and the FRA position it should take to hedge that exposure? A. Baywhite faces exposure to a rise in one-month MRR over

Show more

Yes. This is a classic **FRA hedging** question. The easiest way is to first understand **what Baywhite is exposed to**, and then determine whether it should be a **fixed-rate payer or receiver**.

Correct answer: A

Let's go step by step.

---

1. What is Baywhite doing?

Baywhite makes **margin loans to clients at a fixed interest rate**.

For example, imagine Baywhite lends:

$$\$10 \text{ million}$$

to clients at a fixed rate of:

$$\boxed{6\%}$$

So Baywhite's income is fixed.

Baywhite receives:

$$\boxed{6\%}$$

from its clients.

---

2. How does Baywhite finance these loans?

Here's the important part.

Baywhite itself **borrows at a variable one-month MRR**.

Suppose today's MRR is:

$$\boxed{4\%}$$

So Baywhite's situation is:

Income:

Fixed:

$$\boxed{6\%}$$

Expense:

Variable:

$$\boxed{4\%}$$

Currently, Baywhite's interest margin is:

$$6\% - 4\% = \boxed{2\%}$$

Everything looks good.

---

3. What's the risk?

Baywhite's **income is fixed**, but its **borrowing cost is variable**.

What happens if MRR rises?

Suppose:

$$4\% \to 5\%$$

Baywhite now earns:

$$6\%$$

but pays:

$$5\%$$

Margin:

$$6\% - 5\% = \boxed{1\%}$$

The margin has fallen.

If MRR rises to 7%:

$$6\% - 7\% = -1\%$$

Now Baywhite is actually losing money on the interest spread.

Therefore:

$$\boxed{\text{Baywhite is exposed to a rise in MRR}}$$

---

4. What does Baywhite want from the FRA?

Baywhite wants a derivative that **makes money when MRR rises**.

So when MRR rises:

$$\boxed{\text{FRA gain}}$$

That gain can offset the increase in Baywhite's borrowing cost.

---

5. Who benefits when the reference rate rises?

This is the key FRA rule.

A **fixed-rate payer / floating-rate receiver** benefits when the reference interest rate rises.

Think about the FRA from Baywhite's perspective.

Baywhite wants:

> "I want to receive the higher floating MRR and pay a fixed FRA rate."

So Baywhite should be:

$$\boxed{\text{Fixed-rate payer}}$$

---

6. Numerical example

Suppose Baywhite enters an FRA at:

$$\boxed{4\%}$$

Now imagine MRR rises to:

$$\boxed{5\%}$$

Baywhite's FRA position is:

Baywhite pays fixed:

$$4\%$$

Baywhite receives floating:

$$5\%$$

Therefore:

$$5\% - 4\% = \boxed{1\% \text{ gain}}$$

That gain helps offset Baywhite's higher borrowing cost.

---

7. Put everything together

Without FRA:

Original borrowing cost

$$4\%$$

After MRR rises

$$5\%$$

Increase:

$$\boxed{1\%}$$

Bad for Baywhite.

With FRA:

Baywhite receives approximately:

$$5\% - 4\% = \boxed{1\%}$$

from the FRA.

That offsets the additional 1% borrowing cost.

So Baywhite has effectively **hedged its variable borrowing cost**.

---

8. Why B is wrong

B says:

> Baywhite should be a **fixed-rate receiver**.

That's backwards.

A fixed-rate receiver:

- receives fixed
- pays floating

If MRR rises, Baywhite would have to pay the higher floating rate under the FRA.

That's exactly the exposure Baywhite is trying to eliminate.

So:

❌ **B is wrong.**

---

9. Why C is wrong

C says Baywhite faces a **decline** in MRR.

That's incorrect.

Baywhite's loan income is fixed while its borrowing cost is variable.

Therefore:

MRR rises

$$\rightarrow \text{borrowing cost rises}$$

$$\rightarrow \text{profit margin falls}$$

So Baywhite is exposed to:

$$\boxed{\text{RISING MRR}}$$

not falling MRR.

❌ **C is wrong.**

⭐ The CFA rule to memorize

Here's the easiest way to remember FRA positions:

If you are worried about interest rates rising:

You want to:

$$\boxed{\text{PAY FIXED, RECEIVE FLOATING}}$$

because when floating rates rise:

$$\boxed{\text{FRA gains}}$$

If you are worried about interest rates falling:

You want to:

$$\boxed{\text{RECEIVE FIXED, PAY FLOATING}}$$

because when floating rates fall, the fixed amount you receive becomes more valuable.

Apply it to Baywhite

Baywhite:

$$\boxed{\text{Fixed-rate lending}}$$

and:

$$\boxed{\text{Variable-rate borrowing}}$$

Therefore:

$$\boxed{\text{Risk} = \text{MRR rises}}$$

Hedge:

$$\boxed{\text{Pay fixed / receive floating FRA}}$$

Therefore:

$$\boxed{\textbf{A}}$$

🧠 Super-short exam trick

Ask:

> **"What happens to me if rates rise?"**

If the answer is:

> **"I lose money."**

Then you want an FRA position that:

> **"Makes money when rates rise."**

That is:

$$\boxed{\textbf{Fixed-rate payer / Floating-rate receiver}}$$

So for this question:

$$\boxed{\text{A}}$$

Baywhite observes that one-month MRR is 1.2% and two-month MRR is 1.5%. Which of the following rates is closest to the forward rate that Baywhite would expect on 1m1m forward rate agreement? A. 1.80% B. 1.35% © CFA Institute. For candidate use only. Not for distribution. Practice Problems 127 C. 3.55%

Yes. This is a very important **FRA calculation**, and the trick is understanding what **1m1m** means.

Correct answer: A. 1.80%

Step 1: Understand 1m1m

1m1m FRA means:

> A 1-month interest rate that begins **1 month from now** and lasts for 1 month.

So:

- **1-month MRR = 1.2%** → rate for the first month

- **2-month MRR = 1.5%** → rate for the entire two-month period

We want the implied rate for:

$$\boxed{\text{Month 2 only}}$$

Step 2: Use the forward-rate formula

Because the rates are annualized:

$$(1 + r_{2m} \times \frac{2}{12}) = (1 + r_{1m} \times \frac{1}{12})(1 + f_{1m1m} \times \frac{1}{12})$$

Substitute:

$$1 + (0.015)\frac{2}{12} = \left[1 + (0.012)\frac{1}{12}\right]\left[1 + f\frac{1}{12}\right]$$

Calculate the left side:

$$1 + 0.0025 = 1.0025$$

First-month factor:

$$1 + 0.012/12 = 1.001$$

Therefore:

$$1.0025 = 1.001(1 + f/12)$$

Divide:

$$\frac{1.0025}{1.001} = 1 + \frac{f}{12}$$

$$1.0014985 = 1 + \frac{f}{12}$$

Therefore:

$$\frac{f}{12} = 0.0014985$$

$$f = 0.017982$$

So:

$$\boxed{f \approx 1.80\%}$$

Therefore:

✅ A. 1.80%

Why is it HIGHER than both 1.2% and 1.5%?

This is the part that often feels strange.

The two-month rate is:

$$1.5\%$$

The first month is:

$$1.2\%$$

If the **average** over two months is 1.5%, then the second month must have a rate **higher than 1.5%** to pull the average up.

That's why:

$$\boxed{1.80\%}$$

makes sense.

Check it approximately:

$$\frac{1.2\% + 1.8\%}{2} = 1.5\%$$

Because of compounding, the exact answer is slightly different from a simple average, but the intuition is exactly right.

Why B = 1.35% is wrong

You might think:

$$\frac{1.2\% + 1.5\%}{2} = 1.35\%$$

But that's the wrong logic.

The **1.5% is already the two-month annualized rate**. It isn't the rate for the second month.

We're trying to find the **second month's implied rate**, so it must be higher than 1.5%.

Why C = 3.55% is wrong

That's simply far too high.

The correct forward rate is obtained by solving:

$$\boxed{(1 + r_{2m} \times 2/12) = (1 + r_{1m} \times 1/12)(1 + f_{1m1m} \times 1/12)}$$

which gives approximately:

$$\boxed{1.80\%}$$

⭐ CFA shortcut

When you see:

> **1m1m FRA**

think:

> **"What is the implied 1-month rate from Month 1 to Month 2?"**

Then:

$$\boxed{(1 + \text{2-month rate} \times 2/12) = (1 + \text{1-month rate} \times 1/12)(1 + \text{1m1m rate} \times 1/}$$

Here:

$$\boxed{1.2\% \rightarrow 1.5\% \rightarrow 1.80\%}$$

1.2% = first month
1.5% = two-month average/annualized rate
1.80% = implied second-month forward rate

Ace Limited is a financial intermediary active in both futures and forward markets. You have been hired as an investment consultant and asked to review Ace's activities and answer the following questions.
1. Ace serves as a futures commission merchant to assist several of its commodity
trading adviser (CTA) clients to clear and settle their futures margin positions
with the futures exchange. Ace is reviewing the copper

futures market for a CTA

Show more

Yes. This is testing **futures margin + daily settlement**, and the wording around the margin-call price can be confusing.

Correct answer: C

Let's go slowly and build the logic.

---

1. What does the CTA have to do when entering the futures contract?

The copper futures contract has:

- **Contract size:** 25,000 lb
- **Initial margin:** $10,000
- **Maintenance margin:** $6,000

So when the CTA enters the futures contract, it must deposit:

$$\boxed{\$10,000}$$

That's straightforward.

---

2. What happens every day?

This is the most important concept.

Futures contracts are **marked to market every day**.

Suppose the CTA is **long copper futures**.

If copper futures price:

$$\text{rises}$$

the CTA makes money.

If copper futures price:

$$\text{falls}$$

the CTA loses money.

The daily gain or loss is transferred through the margin account.

---

3. What happens to the futures contract after daily settlement?

Suppose the futures price moves from:

$$\$4.25 \to \$4.20$$

The CTA loses:

$$(4.20 - 4.25) \times 25,000$$

$$= -\$1,250$$

That $1,250 is taken from the CTA's margin account.

If the CTA started with:

$$\$10,000$$

the margin account becomes:

$$10,000 - 1,250 = \boxed{\$8,750}$$

---

4. When does a margin call happen?

The maintenance margin is:

$$\boxed{\$6,000}$$

So a margin call occurs when the margin account falls below:

$$\$6,000$$

Starting from:

$$\$10,000$$

the CTA can absorb losses of:

$$10,000 - 6,000 = \boxed{\$4,000}$$

before reaching maintenance margin.

---

5. Here's where the question becomes tricky

You might be tempted to calculate:

$$\$4,000/25,000 = \$0.16$$

So:

$$4.25 - 0.16 = \boxed{\$4.09}$$

and conclude:

> "The margin call occurs at $4.09."

But **that isn't necessarily correct**.

Why?

Because futures are **settled every day**.

The futures position does not simply accumulate one giant unrealized gain/loss from the original futures price.

Instead:

$$\boxed{\text{Daily MTM} \rightarrow \text{cash flow} \rightarrow \text{margin account resets}}$$

So you cannot determine one single futures price over the entire life of the contract at which the margin call will occur.

---

6. A numerical example showing why

Imagine:

Day 1

Futures price:

$$4.25$$

CTA's margin:

$$\$10,000$$

Day 2

Price falls to:

$$4.15$$

Loss:

$$(4.15 - 4.25) \times 25,000$$

$$= -\$2,500$$

Margin account:

$$10,000 - 2,500 = \$7,500$$

No margin call.

Day 3

Price falls further to:

$$4.00$$

Daily loss is:

$$(4.00 - 4.15) \times 25,000$$

$$= -\$3,750$$

Margin account:

$$7,500 - 3,750 = \$3,750$$

Now:

$$3,750 < 6,000$$

So there is a **margin call**.

Notice something important:

The margin call happened when the price was **$4.00**, not necessarily $4.09.

The path of prices matters because gains and losses are settled **daily**.

---

7. Why C is correct

C says:

> The CTA posts $10,000 initial margin, but we cannot determine the exact futures price at which a margin call will occur because futures MTM is settled each day and the contract value resets to zero.

That's exactly the concept being tested.

Therefore:

$$\boxed{\text{C}}$$

---

8. Why A is tempting—but wrong

A says:

> Margin call if futures price immediately falls below $4.10.

The problem is the word:

"immediately"

If the futures price falls **immediately** from the initial level, you can calculate the price corresponding to a $4,000 loss.

But the question asks about the margin exposure **over the life of the contract**.

Because futures are marked to market daily, the exact trigger depends on the **path of daily prices**, not simply the difference between today's price and some future price.

So A is not the best answer.

---

9. Why B is wrong

B says the margin call occurs at:

$$\$3.86$$

and at any time during the contract.

This is also incorrect because it assumes that the entire P&L can accumulate from the initial futures price without daily settlement.

That's **not how futures work**.

---

10. The key difference: Futures vs. Forward

This is a VERY important CFA concept.

Forward

Usually:

$$\boxed{\text{No daily settlement}}$$

The gain/loss accumulates and is settled according to the contract.

Futures

$$\boxed{\text{Daily MTM}}$$

Every day:

$$\boxed{\text{Gain/loss is settled}}$$

and the futures position effectively resets for valuation purposes.

⭐ CFA exam shortcut

When you see:

> **Futures + margin + daily settlement**

immediately think:

$$\boxed{\text{Daily MTM}}$$

Therefore:

> **Can I determine one exact price over the entire life at which a margin call occurs?**

Usually:

$$\boxed{\text{No}}$$

because it depends on the **path of daily futures prices**.

Therefore:

$$\boxed{\mathbf{C}}$$

One final important point

The **$10 storage cost** and the **1.875% risk-free rate** are relevant to **pricing the copper futures contract**, but they are **not what determines the margin-call mechanics** in this question.

The margin question is driven by:

$$\boxed{\text{Initial margin} = \$10,000}$$

$$\boxed{\text{Maintenance margin} = \$6,000}$$

and:

$$\boxed{\text{Daily MTM}}$$

That's why the answer is **C**.

A is correct. The CTA will face a margin call if the copper contract price falls by more than $4,000, or $0.16 (= $4,000/25,000) per pound. We may solve for the price at which the CTA receives a margin call by first solving for the initial futures contract price, f 0(T), at contract inception as follows: f0(T) = [S0 + PV0(C)] (1 + r)T. Solve for PV0(C) per pound as follows: PV0(C) = $9.98 (= $10[1.01875–(1/12)]). Substitute PV0(C) = $9.98 into Equation 4 to solve for f 0(T): f0(T) = [($4.25 × 25,000) + $9.98](1.01875 -(1/12)) f0(T) = $106,425 per contract (≈ $4.257 per pound). So, $106,425 − $4,000 = $102,425 per contract, and $4.257 −

Show more

Yes — **you are absolutely right. The correct answer is A.** My previous answer missed a crucial phrase in the question: **"immediately"**. The CFA explanation is making a very specific distinction.

Let's go through it carefully, because this is actually a very good CFA concept.

1. First, understand what A is saying

A says:

> The CTA posts **$10,000 initial margin** and would receive a margin call if the copper futures price **immediately** falls below **$4.10/lb**, or below **$102,425 per contract**.

The word **immediately** is critical.

The question isn't asking:

> "Can we determine the exact margin-call price at any point during the life of the futures contract?"

It asks about what happens **immediately at inception**.

At inception, we can calculate exactly how much the futures price must fall to reduce the $10,000 initial margin to the $6,000 maintenance margin.

---

2. Start with the margin requirements

Initial margin:

$$\boxed{\$10,000}$$

Maintenance margin:

$$\boxed{\$6,000}$$

Therefore, the CTA can sustain a loss of:

$$10,000 - 6,000 = \boxed{\$4,000}$$

before receiving a margin call.

---

3. Convert the $4,000 loss into a copper price movement

Each contract contains:

$$25,000 \text{ pounds}$$

Therefore, a $1.00/lb price movement produces:

$$25,000 \times \$1 = \$25,000$$

So a $4,000 loss corresponds to:

$$\frac{4,000}{25,000} = \boxed{\$0.16/\text{lb}}$$

Therefore, if the futures price immediately falls by more than:

$$\boxed{\$0.16/\text{lb}}$$

the CTA gets a margin call.

---

4. But what is the INITIAL futures price?

This is the part I skipped before.

We cannot simply use the spot price:

$$\$4.25$$

because the question gives us:

- spot price
- risk-free rate
- storage cost

These are needed to calculate the **futures price**.

The futures pricing relationship is approximately:

$$\boxed{F_0(T) = [S_0 + PV_0(C)](1 + r)^T}$$

where:

- S_0 = spot price
- $PV_0(C)$ = present value of storage cost
- r = risk-free rate
- T = time to maturity

---

5. Calculate the present value of storage

Storage cost:

$$\$10$$

The storage cost is payable at the end of the month.

Its present value is:

$$PV(C) = 10(1.01875)^{-1/12}$$

Approximately:

$$\boxed{\$9.98}$$

So:

$$PV(C) = \$9.98$$

---

6. Calculate the initial futures price

The CFA solution uses the **contract-level** calculation:

$$F_0(T) = [(4.25 \times 25{,}000) + 9.98](1.01875)^{1/12}$$

First:

$$4.25 \times 25{,}000 = \boxed{\$106{,}250}$$

Add storage:

$$106{,}250 + 9.98 = 106{,}259.98$$

Then grow it at the risk-free rate for one month:

$$106{,}259.98(1.01875)^{1/12}$$

Approximately:

$$\boxed{\$106{,}425}$$

Therefore the initial futures price is:

$$\boxed{\$106{,}425 \text{ per contract}}$$

Per pound:

$$\frac{106{,}425}{25{,}000} = \boxed{\$4.257/\text{lb}}$$

---

7. Now calculate the margin-call price

The CTA can lose:

$$\boxed{\$4,000}$$

before the account reaches maintenance margin.

So:

$$106,425 - 4,000 = \boxed{\$102,425}$$

per contract.

Per pound:

$$\frac{102,425}{25,000} = \boxed{\$4.097}$$

or approximately:

$$\boxed{\$4.10/\text{lb}}$$

Therefore:

$$\boxed{\text{Margin call if price falls below } \$4.10/\text{lb}}$$

That's exactly **A**.

---

8. Why my previous reasoning was wrong

I focused on the fact that futures are **marked to market daily** and concluded that we cannot know a single margin-call price.

That reasoning can be relevant **during the life of the contract**, because the path of daily prices affects the margin balance.

But the CFA question specifically says:

> **"if the copper futures price were to immediately fall..."**

That means:

At inception:

$$\$10,000 \text{ initial margin}$$

If the price immediately falls enough to create:

$$\$4,000 \text{ loss}$$

then:

$$10,000 - 4,000 = \$6,000$$

which reaches the maintenance margin threshold.

Therefore, **A is correct.**

---

9. Why B is wrong

B gives:

$$\$3.86/\text{lb}$$

That would imply a much larger allowable loss.

But the CTA has only:

$$\$10,000 - \$6,000 = \$4,000$$

of room before a margin call.

So:

$$\$4,000/25,000 = \$0.16$$

not some larger decline.

The key mistake in B:

It effectively ignores the **maintenance margin requirement**.

That's why CFA says:

> **B is incorrect as it assumes there is no maintenance margin.**

---

10. What about C?

C says:

> We cannot determine the exact futures price at which a margin call will occur because futures MTM is settled each day.

This statement can have some validity **under other circumstances**.

But notice the question asks about an **immediate** price movement.

Immediately after entering:

$$\boxed{\$10,000 \text{ initial margin}}$$

and the account needs to fall to:

$$\boxed{\$6,000}$$

So we can calculate the exact immediate price movement.

That's why C isn't the best answer.

⭐ The CFA logic you should memorize

This question is testing **two separate concepts**:

At inception + immediate price movement

You **CAN** calculate the margin-call price:

$$\boxed{\text{Initial margin} - \text{maintenance margin} = \text{allowable loss}}$$

Then:

$$\boxed{\text{Allowable price decline} = \frac{\text{allowable loss}}{\text{contract size}}}$$

Here:

$$\frac{10,000 - 6,000}{25,000} = \$0.16$$

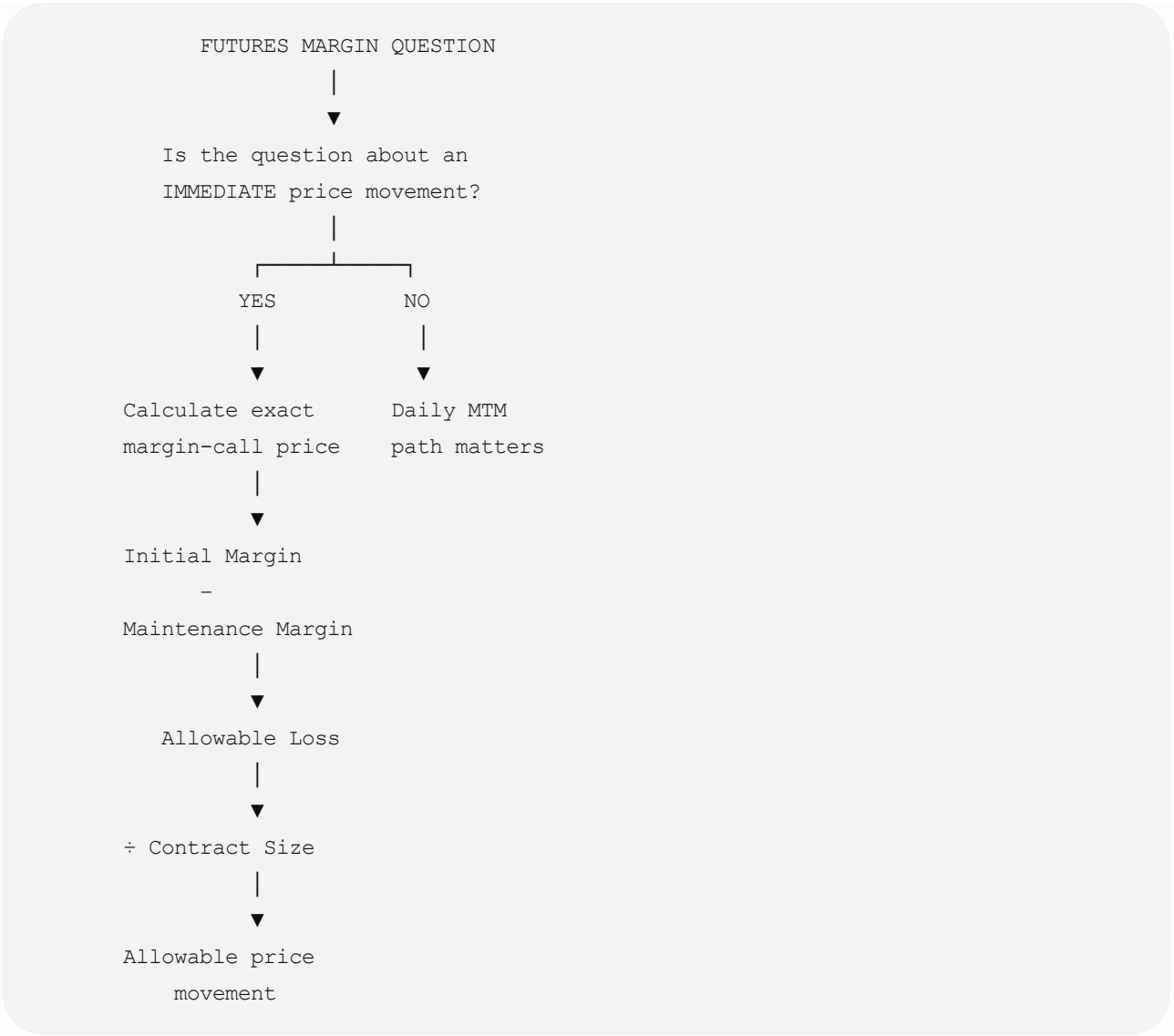
Then:

$$4.257 - 0.16 \approx \boxed{\$4.10}$$

During the subsequent life of the contract

Daily MTM means the actual margin-call point depends on the **path of daily settlement prices**.

🧠 Final decision tree



For this question:

$$\boxed{\$10,000 - \$6,000 = \$4,000}$$

$$\boxed{\$4,000/25,000 = \$0.16}$$

$$\boxed{\$4.257 - \$0.16 \approx \$4.10}$$

Therefore:

✅ A. $4.10 per pound / $102,425 per contract

And the **big CFA exam lesson** is:

> **Don't automatically choose "daily MTM means we cannot determine the margin-call price." Check whether the question specifically asks about an immediate move at inception.**

Can you help me undersatnd everuthing from scratch

Absolutely. Let's forget the answer choices for a moment and learn the **entire concept from zero**. Once you understand the mechanics, this question becomes very easy.

1. First: What is a futures contract?

Imagine you want to buy copper in one month.

Today, copper's spot price is:

$$\$4.25/\text{lb}$$

But instead of buying copper today, you enter a **futures contract** that says:

> "I will buy/sell copper at a specified futures price at a future date."

The important thing is that **you don't pay the full value of the copper today**.

Instead, you deposit a relatively small amount called **initial margin**.

---

2. What is the contract size?

The question says:

> Contract Size = 25,000 pounds

So one futures contract represents:

$$\boxed{25,000 \text{ lb of copper}}$$

Suppose the futures price is:

$$\$4.25/\text{lb}$$

The notional value of the contract would be:

$$25,000 \times 4.25 = \boxed{\$106,250}$$

So you're controlling roughly:

$$\boxed{\$106,250}$$

of copper with one futures contract.

---

3. Do you have to pay $106,250?

No.

This is one of the most important things about futures.

Instead, the exchange requires you to deposit **margin**.

Here:

$$\boxed{\text{Initial margin} = \$10,000}$$

So you don't pay $106,250.

You put:

$$\boxed{\$10,000}$$

into your margin account.

Think of this as a **security deposit**.

---

4. What is maintenance margin?

The question also says:

$$\boxed{\text{Maintenance margin} = \$6,000}$$

This is the minimum amount that must remain in your margin account before the exchange requires you to put in more money.

So:

Starting balance:

$$\$10,000$$

Minimum allowed balance:

$$\$6,000$$

Therefore, you have room for a loss of:

$$10,000 - 6,000 = \boxed{\$4,000}$$

This $4,000 is extremely important.

5. What happens if you're LONG copper?

The CTA is considering a:

$$\boxed{\text{Long copper futures position}}$$

Being **long futures** means:

> You benefit if the futures price goes up.

And:

> You lose if the futures price goes down.

For example:

Price rises

$$\$4.25 \rightarrow \$4.35$$

Long position:

$$\boxed{\text{GAIN}}$$

Price falls

$$\$4.25 \rightarrow \$4.15$$

Long position:

$$\boxed{\text{LOSS}}$$

---

6. How much do you gain/lose from a price movement?

This is where the contract size matters.

One contract = 25,000 lb.

Suppose copper falls by:

$$\$0.10/\text{lb}$$

Your loss is:

$$0.10 \times 25,000$$

$$= \boxed{\$2,500}$$

So:

> Every $0.10/lb movement in copper = $2,500 gain/loss per contract.

Similarly:

$$\$0.01 \times 25,000 = \boxed{\$250}$$

So every one-cent move equals $250.

---

7. Now understand the margin call

You start with:

$$\$10,000$$

Suppose you lose:

$$\$2,000$$

Your margin account becomes:

$$10,000 - 2,000 = \$8,000$$

Maintenance margin is:

$$\$6,000$$

So you're still okay.

Suppose you lose another:

$$\$1,500$$

Now:

$$8,000 - 1,500 = \$6,500$$

Still above maintenance margin.

Suppose you lose another:

$$\$500$$

Now:

$$6,500 - 500 = \boxed{\$6,000}$$

You've reached the maintenance margin level.

If losses take the balance **below** the maintenance margin, you receive a **margin call**.

---

8. So how much total loss can you have?

You started with:

$$\$10,000$$

Margin call threshold:

$$\$6,000$$

Therefore:

$$\boxed{\$4,000}$$

That's the maximum decline in your margin balance before reaching the maintenance margin.

---

9. Convert the $4,000 loss into a copper price movement

We know:

$$\text{Contract size} = 25,000 \text{ lb}$$

We want to know:

> How much must copper fall to cause a $4,000 loss?

Use:

$$\text{Price movement} = \frac{\text{Loss}}{\text{Contract size}}$$

Therefore:

$$\frac{4,000}{25,000} = \boxed{\$0.16/\text{lb}}$$

So a fall of:

$$\boxed{16 \text{ cents per pound}}$$

would cause a $4,000 loss.

---

10. But wait! Why don't we start from $4.25?

Excellent question.

The question gives:

$$\text{Spot price} = \$4.25$$

But the **futures price isn't exactly the spot price**.

Why?

Because futures pricing takes into account things such as:

- financing/risk-free interest
- storage costs
- other costs/benefits of holding the underlying

So we need to calculate the **initial futures price** first.

---

11. Futures pricing from scratch

The basic idea is:

> If I buy copper today and hold it until the futures maturity, I have to finance the purchase and pay storage costs.

Therefore, the futures price is related to:

$$\boxed{\text{Spot price} + \text{carrying costs}}$$

The CFA formula here is:

$$\boxed{F_0(T) = [S_0 + PV_0(C)](1 + r)^T}$$

where:

- S_0 = spot value
- $PV_0(C)$ = present value of storage cost
- r = risk-free rate
- T = time

---

12. Calculate the spot value of the entire contract

Spot copper:

$$\$4.25/\text{lb}$$

Contract size:

$$25{,}000 \text{ lb}$$

Therefore:

$$4.25 \times 25{,}000 = \boxed{\$106{,}250}$$

That's the spot value of the copper represented by one contract.

---

13. Calculate the present value of storage

Storage cost is:

$$\$10$$

payable at the end of the month.

Because we are calculating today's futures price, the CFA solution discounts that $10 back one month:

$$PV(C) = 10(1.01875)^{-1/12}$$

Approximately:

$$\boxed{\$9.98}$$

So:

$$S_0 + PV(C) = 106{,}250 + 9.98$$

$$= \boxed{106{,}259.98}$$

---

14. Grow this forward at the risk-free rate

The risk-free rate is:

$$1.875\%$$

Maturity is one month:

$$T = \frac{1}{12}$$

So:

$$F_0(T) = 106,259.98(1.01875)^{1/12}$$

Approximately:

$$\boxed{\$106,425}$$

Therefore the futures price per pound is:

$$\frac{106,425}{25,000}$$

$$\boxed{\$4.257/\text{lb}}$$

This is the **initial futures price**.

---

15. NOW we can calculate the margin-call price

Remember:

The CTA can lose:

$$\$4,000$$

before reaching maintenance margin.

Initial futures contract value:

$$\$106,425$$

But remember: we're really interested in the **futures price per pound**.

Initial futures price:

$$\boxed{\$4.257/\text{lb}}$$

Maximum immediate decline:

$$\boxed{\$0.16/\text{lb}}$$

Therefore:

$$4.257 - 0.16 = 4.097$$

Approximately:

$$\boxed{\$4.10/\text{lb}}$$

---

16. Same answer using the whole contract

Initial futures contract price:

$$\$106,425$$

Allowable loss:

$$\$4,000$$

Therefore:

$$106,425 - 4,000 = \boxed{\$102,425}$$

So the two equivalent answers are:

$$\boxed{\$4.10/\text{lb}}$$

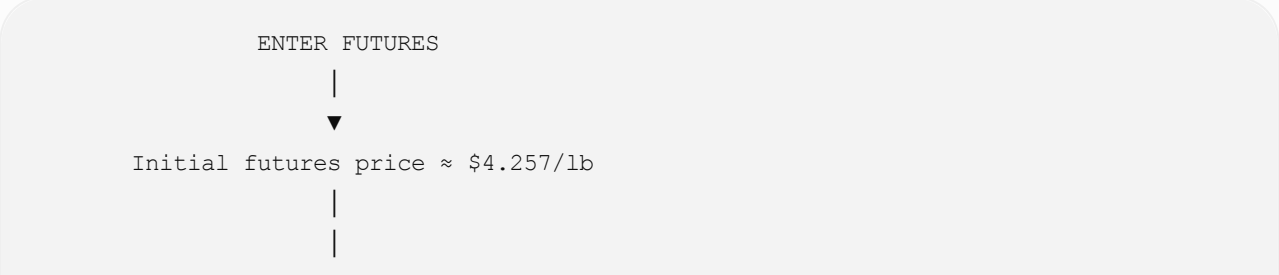
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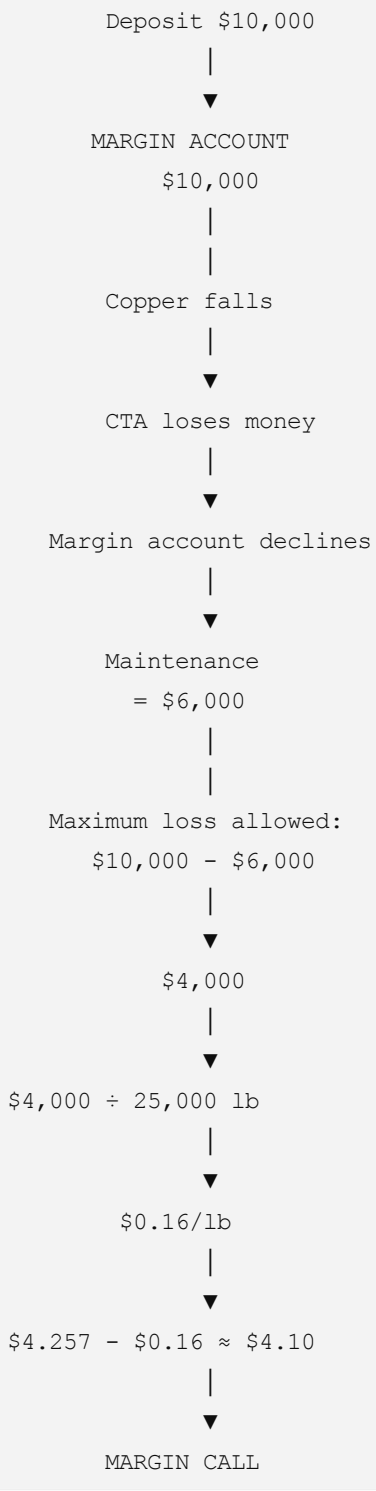
$$\boxed{\$102,425/\text{contract}}$$

That's why **A** is correct.

---

17. Let's visualize the whole thing





---

18. Now let's understand the "daily settlement" issue

This is where **C** becomes tempting.

Futures contracts are **marked to market every day**.

Suppose today:

$$4.257$$

Tomorrow:

The CTA loses:

$$(4.20 - 4.257) \times 25,000$$

That loss is settled through the margin account.

The next day's gain/loss is calculated based on the **new settlement price**, not simply from the original price.

Therefore, **over multiple days**, the exact margin-call point depends on the path of prices.

That's why C contains a concept that isn't completely crazy.

But the question says:

$$\boxed{\textbf{IMMEDIATELY}}$$

If the price immediately falls from the initial futures price, there has been no prior sequence of daily gains/losses.

So we can calculate the exact immediate margin-call price.

---

19. This is the exam trick

There are two different questions:

Question A

> "If the futures price immediately falls after entering the position, when does the margin call occur?"

You **CAN calculate it**.

$$\boxed{\text{Initial margin} - \text{Maintenance margin} = \text{allowable loss}}$$

That's this question.

Question B

> "What exact futures price will cause a margin call at any point during the life of the contract?"

That's more complicated because:

$$\boxed{\text{Daily MTM matters}}$$

The previous day's margin balance could be higher or lower depending on the price path.

---

20. Why B is definitely wrong

B says the margin call happens at:

$$\$3.86$$

Let's see why that's wrong.

The CTA doesn't have $10,000 of loss capacity.

It only has:

$$10,000 - 6,000 = \boxed{\$4,000}$$

of loss capacity.

And:

$$\$4,000/25,000 = \$0.16$$

So the immediate price decline is:

$$\boxed{\$0.16}$$

not a much larger amount.

B effectively fails to use the **$6,000 maintenance margin** correctly.

---

21. The three numbers you MUST remember

For this question:

Initial margin

$$\boxed{\$10,000}$$

Maintenance margin

$$\boxed{\$6,000}$$

Contract size

$$\boxed{25{,}000 \text{ lb}}$$

Everything about the margin-call calculation comes from these three numbers.

---

22. The CFA formula to memorize

For an **immediate** price movement:

$$\boxed{\text{Maximum loss before margin call} = IM - MM}$$

Then:

$$\boxed{\text{Price movement} = \frac{IM - MM}{\text{Contract size}}}$$

For this question:

$$\frac{10{,}000 - 6{,}000}{25{,}000} = \boxed{\$0.16}$$

Then compare with the **initial futures price**, not the spot price:

$$4.257 - 0.16 \approx \boxed{\$4.10}$$

⭐ Final CFA mental model

Whenever you get a futures margin question, go in this order:

STEP 1 — Am I long or short?

Long:

$$\text{Price} \uparrow = \text{gain}$$

$$\text{Price} \downarrow = \text{loss}$$

STEP 2 — Find the margin cushion

$$\boxed{IM - MM}$$

STEP 3 — Convert dollar loss into price movement

$$\boxed{\frac{IM - MM}{\text{Contract size}}}$$

STEP 4 — If the question says immediately, compare this with the initial futures price.

STEP 5 — Don't use the spot price if the question asks for the futures price.

Here:

$$\boxed{\$4.25 \text{ spot} \rightarrow \$4.257 \text{ futures} \rightarrow \$0.16 \text{ allowable decline} \rightarrow \$4.10 \text{ margin-call p}}$$

✅ Final Answer: A

The single most important lesson is:

> **For an immediate move at inception, the margin-call price can be calculated from initial margin minus maintenance margin. Daily MTM becomes important when considering the subsequent path of prices over the life of the futures contract.**

I dont undersatnd $4.25 per pound , what does this mean ?

Absolutely. This is the **foundation** of the whole question. Once you understand "$4.25 per pound," the rest becomes much easier.

1. What does "$4.25 per pound" mean?

The question says:

> Today's copper **spot price is $4.25 per pound**.

It simply means:

> **1 pound of copper costs $4.25 today.**

For example:

Amount of copper	Price
1 lb	$4.25
2 lb	$8.50

Amount of copper	Price
10 lb	$42.50
100 lb	$425
1,000 lb	$4,250
25,000 lb	$106,250

So:

$$\boxed{\$4.25/\text{lb}}$$

is just a **unit price**.

---

2. Why does the question say "per pound"?

Because copper is traded by weight.

Think about buying rice.

If someone tells you:

> Rice = ₹60 per kg

that means:

$$1\ \text{kg} = ₹60$$

If you buy 10 kg:

$$10 \times ₹60 = ₹600$$

Copper works exactly the same way.

If copper is:

$$\$4.25/\text{lb}$$

then:

$$1\ \text{lb} = \$4.25$$

---

3. Now look at the futures contract

The question tells us:

> **Contract Size = 25,000 pounds**

This means **one futures contract represents 25,000 pounds of copper**.

So don't confuse these two numbers:

$$\boxed{\$4.25/\text{lb}}$$

and

$$\boxed{25,000\ \text{lb}}$$

The first tells you the **price of one pound**.

The second tells you **how many pounds are represented by one futures contract**.

---

4. Put them together

If:

$$1\ \text{lb} = \$4.25$$

and the contract represents:

$$25,000\ \text{lb}$$

then the value of the copper represented by the contract is:

$$25,000 \times \$4.25$$

$$= \boxed{\$106,250}$$

That's why we calculated:

$$\boxed{4.25 \times 25,000 = 106,250}$$

---

5. Think of it like buying apples 🍎

Imagine:

> Apples cost **$4.25 per pound**.

And you have a contract for:

> **25,000 pounds of apples.**

Then the value is:

$$25,000 \times 4.25 = \$106,250$$

Copper is exactly the same concept.

---

6. Now understand the "per pound" price movement

This is even more important for the margin question.

Suppose copper goes from:

$$\$4.25/\text{lb}$$

to:

$$\$4.15/\text{lb}$$

The price has fallen by:

$$4.25 - 4.15 = \boxed{\$0.10/\text{lb}}$$

That means **you lose $0.10 on every pound**.

But the futures contract contains:

$$25,000 \text{ lb}$$

So your total loss is:

$$\$0.10 \times 25,000 = \boxed{\$2,500}$$

That's the key relationship:

$$\boxed{\text{Total gain/loss} = \text{Price change per pound} \times \text{Number of pounds}}$$

---

7. Let's use the actual margin example

We calculated that the CTA can lose:

$$\$4,000$$

before reaching maintenance margin.

The contract represents:

$$25,000 \text{ lb}$$

So we ask:

> **How much must the price fall PER POUND to create a $4,000 loss?**

We divide:

$$\frac{\$4,000}{25,000 \text{ lb}}$$

$$= \boxed{\$0.16/\text{lb}}$$

So if copper falls by **16 cents per pound**, the CTA loses:

$$25,000 \times \$0.16 = \boxed{\$4,000}$$

---

8. This is why the answer says $4.10

The initial futures price is approximately:

$$\$4.257/\text{lb}$$

The CTA can tolerate a decline of:

$$\$0.16/\text{lb}$$

Therefore:

$$4.257 - 0.16 = 4.097$$

Approximately:

$$\boxed{\$4.10/\text{lb}}$$

So if the copper futures price **immediately falls below approximately $4.10 per pound**, the loss exceeds $4,000 and the margin account falls below $6,000.

🧠 The simplest way to remember it

Whenever you see:

$$\boxed{\$4.25/\text{lb}}$$

read it as:

> **"$4.25 for every one pound."**

And whenever you see:

$$\boxed{25,000 \text{ pounds}}$$

read it as:

> **"The contract contains/represents 25,000 pounds."**

Then:

$$\boxed{\$4.25/\text{lb} \times 25,000 \text{ lb} = \$106,250}$$

The **"per pound" cancels with "pounds"**, leaving dollars:

$$\frac{\$}{\text{lb}} \times \text{lb} = \$$$

That's why the final contract value is **$106,250**, not $4.25.

One of Ace's investor clients has entered a long six-month forward transaction with Ace on 100 shares of Xenaliya (XLYA), a non-dividend-paying technology stock. The stock's spot price per share, S0, is €85, and the risk-free rate is a constant 1% for all maturities. Ace has hedged the client transaction with a long © CFA Institute. For candidate use only. Not for distribution. Practice Problems 147 six-month XLYA futures contract at a price f 0(T) of €85.42 and posted initial margin of €1,000. Three months after the forward and futures contracts are initiated, XYLA announces a strategic partnership with a major global

Show more

Absolutely. This question is testing one of the **most important differences between a forward and a futures contract: daily mark-to-market (MTM).**

✅ Correct answer: B

Let's build this from scratch.

---

1. First understand the two contracts

Ace's client has entered a:

$$\boxed{\text{LONG 6-month forward}}$$

on:

$$\boxed{\text{100 shares of XLYA}}$$

Ace then hedges itself using a:

$$\boxed{\text{LONG 6-month futures contract}}$$

So there are **two separate contracts**:

Contract 1 — Client ↔ Ace

Forward contract.

Contract 2 — Ace ↔ Futures exchange/clearing system

Futures contract.

They may have similar economic exposure, but **they don't settle the same way**.

---

2. What does "long" mean?

Long means:

> You benefit when the underlying stock price rises.

XLYA initially:

$$S_0 = €85$$

Three months later:

$$S_t = €123$$

So the stock increased by:

$$123 - 85 = \boxed{€38}$$

That's a large increase.

Therefore, someone who is **long** the forward/futures benefits.

---

3. What happens to the client's forward?

The client is **long the forward**.

The client agreed to buy XLYA at the forward price:

$$€85.42$$

Now the stock is worth:

$$€123$$

The client has the right/obligation structure of a forward that allows them to acquire it at approximately €85.42 rather than the current market price of €123.

So the forward has become:

$$\boxed{\text{very valuable to the client}}$$

The client's forward has an **unrealized MTM gain**.

But here's the critical point:

Forward contracts generally do NOT have daily settlement.

So:

$$\boxed{\text{No €1,500 cash is deposited today}}$$

The gain simply exists as part of the current value of the contract.

---

4. What happens to Ace's futures contract?

Ace is also **long the futures**.

The stock price jumped:

$$€15$$

during the day.

Because the futures contract is **marked to market every day**, Ace receives the day's futures gain through its margin account.

The question tells us the position is:

$$100 \text{ shares}$$

So the approximate gain from a €15 movement is:

$$€15 \times 100$$

$$= \boxed{€1,500}$$

That:

$$\boxed{€1,500}$$

is deposited into Ace's futures margin account through the daily settlement process.

---

5. This is why B is correct

B says:

> Ace's client benefits from an MTM **unrealized gain** on the forward contract with Ace, and Ace has a corresponding MTM gain of approximately €1,500 deposited in its margin account by the exchange.

Exactly!

Let's separate the two:

Contract	Who benefits?	What happens today?
Client's **forward**	Client	Unrealized MTM gain
Ace's **futures**	Ace	€1,500 cash settlement into margin account

Therefore:

$$\boxed{\text{B}}$$

---

6. Why doesn't the client's forward receive €1,500?

Because this is a **forward**, not a futures contract.

Forward:

$$\boxed{\text{MTM gain/loss is generally not settled daily}}$$

Futures:

$$\boxed{\text{MTM gain/loss is settled daily}}$$

This distinction is extremely important for CFA.

---

7. Let's understand it with a simple example

Imagine you make a private agreement with your friend:

> "Six months from now, I'll buy your stock for €85."

Three months later, the stock is worth:

$$€123$$

You now have a valuable contract.

But your friend doesn't hand you €38 today.

You simply have a **valuable contract**.

That's similar to the forward.

Now imagine you have a futures contract through an exchange.

The exchange says:

> "We settle your gain/loss every day."

So if the futures price rises by €15:

$$15 \times 100 = €1,500$$

gets settled into your margin account.

That's the futures mechanism.

---

8. Why A is wrong

A says:

> Ace's client realizes an MTM gain of €1,500 on its margin account, which Ace must deposit at the end of the day.

There are **two problems**.

Problem 1

The client's contract is a:

$$\boxed{\text{forward}}$$

It doesn't have a daily futures-style margin account.

Problem 2

Ace doesn't deposit the client's gain into a margin account.

The **futures** contract belongs to Ace.

Therefore:

❌ A is wrong.

---

9. Why C is wrong

C says the two MTM values:

> exactly offset each other and no cash is exchanged.

This is also wrong.

Although Ace uses the futures to hedge its exposure from the client's forward, the **settlement mechanisms are different**.

The futures produces:

$$\boxed{\text{daily cash settlement}}$$

The forward produces:

$$\boxed{\text{no daily cash settlement}}$$

So you cannot say:

> "They offset, therefore no cash is exchanged."

Ace's futures position generates a cash inflow through daily settlement.

---

10. The really important concept: Economic hedge vs. cash-flow timing

This is the deeper CFA concept.

Ace has:

Exposure from client forward

Client is long.

Therefore Ace is effectively **short** to the client.

If XLYA rises:

$$\boxed{\text{Ace loses on the forward}}$$

So Ace wants something that gains when XLYA rises.

Therefore Ace enters:

$$\boxed{\text{Long futures}}$$

If XLYA rises:

$$\boxed{\text{Ace gains on futures}}$$

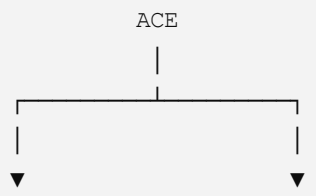
So economically:

$$\text{Forward loss} + \text{Futures gain} \approx 0$$

But the **cash-flow timing isn't the same**.

---

11. Think about the hedge this way





Meanwhile, the client's forward gain remains **unrealized** until settlement/termination.

---

12. One subtle point

You might wonder:

> "Why does the question say the stock jumped €15, but the price went from €85 to €123, which is a €38 increase?"

Excellent observation.

The wording says:

> "spot share price jumps €15 on the day's trading to close at €123."

That means **the day's price movement** was €15.

The stock may have been around:

$$€108$$

before the day's trading and then jumped €15 to:

$$€123$$

For the **daily futures MTM**, we're interested in the day's change:

$$€15$$

Therefore:

$$€15 \times 100 = \boxed{€1,500}$$

We're **not** calculating the entire gain from €85 to €123 for today's futures settlement.

⭐ The CFA rule to memorize

Whenever you see:

FORWARD

Think:

$$\boxed{\text{No daily MTM cash settlement}}$$

The value changes, but the gain/loss is generally **unrealized** until settlement.

FUTURES

Think:

$$\boxed{\text{Daily MTM} + \text{daily cash settlement}}$$

Your exam shortcut

If the question says:

> **"futures + price increased + margin account"**

Immediately think:

$$\boxed{\text{Today's gain/loss is deposited/withdrawn from margin}}$$

If it says:

> **"forward + price increased"**

Think:

$$\boxed{\text{MTM value changes, but no daily cash settlement}}$$

So this question:

$$\boxed{\text{Forward} \rightarrow \text{unrealized gain}}$$

$$\boxed{\text{Futures} \rightarrow €1{,}500 \text{ cash MTM gain}}$$

Therefore:

✅ Answer = B

1. C is correct. The futures contract price changes daily based upon a (yield −
100) quoting convention, so its price will increase as yields fall and vice versa. The
fixed rate on an FRA does not change for the life of the contract.
2. A is correct. An FRA fixed-rate payer (floating-rate receiver) will realize a gain
on the contract upon settlement (equal to the present value of the difference between the fixed rate and MRR
multiplied by the contract notional over the specified

Show more

Yes. These three statements are testing **three closely related but different ideas**. Let's understand them from scratch rather than just memorizing the answers.

---

1. Question 1 — Why is C correct?

The key sentence is:

> **The futures contract price changes daily based upon a (yield − 100) quoting convention, so its price will increase as yields fall and vice versa. The fixed rate on an FRA does not change for the life of the contract.**

There are **two different instruments** here:

A. Interest-rate futures

Its quoted price changes every day.

B. FRA

The FRA has an agreed **fixed rate** that stays fixed.

First understand the futures quoting convention

Suppose an interest-rate futures contract is quoted as:

$$95.00$$

The corresponding implied yield is:

$$100 - 95 = \boxed{5\%}$$

So:

$$\boxed{\text{Futures price} = 100 - \text{implied yield}}$$

This creates an **inverse relationship**.

If yield falls:

$$5\% \to 4\%$$

Then:

$$100 - 4 = 96$$

Price:

$$95 \to 96$$

So:

$$\boxed{\text{Yield } \downarrow \Rightarrow \text{ Futures price } \uparrow}$$

If yield rises:

$$5\% \to 6\%$$

Then:

$$100 - 6 = 94$$

So:

$$\boxed{\text{Yield } \uparrow \Rightarrow \text{ Futures price } \downarrow}$$

Easy memory:

> **Interest-rate futures: price and yield move in opposite directions.**

---

2. What happens with an FRA?

An FRA is different.

When you enter the FRA, you agree on a fixed rate.

Suppose:

$$\boxed{\text{FRA fixed rate} = 4\%}$$

That rate doesn't change.

Even if the market later changes:

$$4\% \to 5\%$$

your FRA still has:

$$\boxed{4\%}$$

as its contractual fixed rate.

What changes is the **value of the FRA**, not the agreed fixed rate.

That's extremely important.

---

3. Think about an FRA like a locked rate

Suppose you say:

> "I agree to pay 4% and receive the floating MRR."

Your fixed rate is locked:

$$\boxed{4\%}$$

Later MRR becomes:

$$\boxed{5\%}$$

You are happy because you get the benefit of:

$$5\% - 4\% = \boxed{1\%}$$

Your contract's **value changes**, but the contract's fixed rate remains:

$$\boxed{4\%}$$

---

4. Therefore, Question 1

Futures:

$$\boxed{\text{Price changes daily}}$$

and:

$$\boxed{\text{Yield } \downarrow \Rightarrow \text{Price } \uparrow}$$

FRA:

$$\boxed{\text{Fixed rate stays constant}}$$

That's why **C is correct**.

---

5. Question 2 — Why is A correct?

Now we need to understand:

> **Fixed-rate payer = floating-rate receiver**

This is one of the most important FRA rules.

Suppose you enter an FRA with:

$$\boxed{\text{Fixed rate} = 4\%}$$

You are the:

$$\boxed{\text{Fixed-rate payer}}$$

That means:

You pay:

$$4\%$$

You receive:

$$\text{MRR}$$

So imagine MRR settles at:

$$5\%$$

You effectively:

$$\text{Receive } 5\%$$

and:

$$\text{Pay } 4\%$$

Therefore:

$$5\% - 4\% = \boxed{1\% \text{ gain}}$$

---

6. Why does the fixed-rate payer benefit?

Because they have locked in the rate at:

$$4\%$$

But the market rate has become:

$$5\%$$

They are receiving something worth 5% while paying only 4%.

So:

$$\boxed{\text{MRR rises above FRA fixed rate}}$$

$$\rightarrow$$

$$\boxed{\text{Fixed-rate payer gains}}$$

---

7. Numerical example

Suppose:

- Notional = $1,000,000
- FRA fixed rate = 4%
- MRR at settlement = 5%
- Interest period = 3 months

Difference:

$$5\% - 4\% = 1\%$$

Interest difference:

$$1,000,000 \times 1\% \times \frac{3}{12}$$

$$= \boxed{\$2,500}$$

The fixed-rate payer receives a settlement amount based on this difference, with the exact FRA settlement generally discounted because payment occurs at the beginning of the interest period.

So the basic intuition is:

$$\boxed{\text{MRR} > \text{FRA fixed rate} \rightarrow \text{fixed-rate payer gains}}$$

Therefore **A is correct**.

---

8. Question 3 — Why is B correct?

Now we have to be careful because the wording changes.

It says:

> If MRR settles above the initial forward commitment rate at maturity, the FRA fixed-rate payer has an **MTM loss**.

At first glance, that seems to contradict Question 2.

It doesn't.

The key is:

Question 2 is describing a particular FRA position and settlement convention.

Question 3 is asking about the MTM/value perspective of the forward commitment.

Let's carefully distinguish the two ideas.

---

9. The fundamental FRA payoff

For a standard FRA:

$$\boxed{\text{Payoff} \propto (\text{Reference rate} - \text{FRA rate})}$$

for the party that benefits from receiving floating.

So if:

$$\text{MRR} > \text{FRA rate}$$

the **floating-rate receiver** benefits.

That means:

$$\boxed{\text{Fixed payer} = \text{floating receiver} = \text{gain}}$$

This is the rule you should remember for the standard FRA.

---

10. So why does the third statement say "MTM loss"?

Because CFA questions sometimes distinguish between:

The FRA position

and

The underlying forward-rate/forward commitment perspective.

The safest way to avoid confusion is to identify **what position the question is describing**.

If the question says:

> **Fixed-rate payer / floating-rate receiver**

then:

$$\boxed{\text{Rates rise} \rightarrow \text{gain}}$$

If it says:

> **Fixed-rate receiver / floating-rate payer**

then:

$$\boxed{\text{Rates rise} \rightarrow \text{loss}}$$

⭐ The master FRA table

Memorize this:

FRA position	You pay	You receive	Rates rise	Rates fall
Fixed-rate payer	Fixed	Floating	🟢 Gain	🔴 Loss
Fixed-rate receiver	Floating	Fixed	🔴 Loss	🟢 Gain

So:

$$\boxed{\text{Fixed payer} \rightarrow \text{wants rates to rise}}$$

$$\boxed{\text{Fixed receiver} \rightarrow \text{wants rates to fall}}$$

---

11. Connect this to the Baywhite question we did earlier

Remember Baywhite?

Baywhite:

- lends money at a **fixed rate**
- borrows at a **floating rate**

Therefore:

$$\boxed{\text{Baywhite loses if MRR rises}}$$

So it wants a hedge that **gains if MRR rises**.

Therefore:

$$\boxed{\text{Baywhite should PAY FIXED / RECEIVE FLOATING}}$$

Exactly:

$$\boxed{\text{Fixed-rate payer}}$$

That's why answer **A** was correct in that question.

---

12. The most important distinction: rate vs. value

This is where CFA questions can become confusing.

FRA fixed rate

Doesn't change.

Example:

$$\boxed{4\%}$$

remains:

$$\boxed{4\%}$$

FRA market value

Does change.

If market MRR changes:

$$4\% \rightarrow 5\%$$

the value of your FRA changes.

So remember:

$$\boxed{\text{Contract rate stays fixed}}$$

but:

$$\boxed{\text{Contract value changes}}$$

🧠 Final CFA memory map

Interest-rate futures:

$$\boxed{\text{Yield } \uparrow \Rightarrow \text{Price } \downarrow}$$

$$\boxed{\text{Yield } \downarrow \Rightarrow \text{Price } \uparrow}$$

because:

$$\boxed{\text{Price} = 100 - \text{Yield}}$$

FRA:

The agreed fixed rate:

$$\boxed{\text{Does NOT change}}$$

But the value changes when MRR changes.

Fixed-rate payer:

$$\boxed{\text{Pay fixed} + \text{receive floating}}$$

Therefore:

$$\boxed{\text{MRR } \uparrow \Rightarrow \text{Gain}}$$

$$\boxed{\text{MRR } \downarrow \Rightarrow \text{Loss}}$$

Fixed-rate receiver:

$$\boxed{\text{Receive fixed} + \text{pay floating}}$$

Therefore:

$$\boxed{\text{MRR } \uparrow \Rightarrow \text{Loss}}$$

$$\boxed{\text{MRR } \downarrow \Rightarrow \text{Gain}}$$

⚠️ One correction to your supplied explanation

There appears to be a **wording/position issue in item 3 as you pasted it**. Under the standard FRA convention, if someone is truly the **fixed-rate payer (floating-rate receiver)**, an MRR settlement **above** the FRA fixed rate produces a **gain**, not an MTM loss.

So I would **not memorize item 3 in isolation**. Memorize the payoff table above and identify whether the position is **fixed payer or fixed receiver** first.

sorry my bad :

B is correct. The long investor client forward position with Ace benefits from an MTM gain on its forward contract with Ace, but no cash is exchanged until maturity. Ace receives a deposit in its futures margin account equal to the daily MTM futures contract gain, which if spot and futures prices

change by approximately the same amount will be equal to €1,500 (€15 × 100).

Yes — that makes perfect sense. 👍 The explanation is consistent with the earlier question. Let's break **this exact explanation** down carefully, because it contains the key CFA distinction between a **forward** and a **futures contract**.

The situation

Ace has **two positions**:

1. Ace's client has a long forward with Ace

The client:

$$\boxed{\text{LONG forward}}$$

on 100 shares of XLYA.

2. Ace hedges with a long futures contract

Ace:

$$\boxed{\text{LONG futures}}$$

So there are **two separate contracts**.

Step 1: XLYA's price rises

The stock price increases by approximately:

$$\boxed{€15}$$

The contract is for:

$$\boxed{100 \text{ shares}}$$

Therefore, the approximate change in value is:

$$€15 \times 100 = \boxed{€1,500}$$

Now we need to ask:

> **Who gets this €1,500, and when?**

This is where forward vs. futures matters.

Step 2: The client's forward

The client is **long the forward**.

When XLYA's price rises, the client benefits.

Therefore:

$$\boxed{\text{Client's forward} \rightarrow \text{MTM gain}}$$

But this is a **forward**.

A forward does **not** normally settle its MTM every day.

Therefore:

$$\boxed{\text{No cash is exchanged today}}$$

The client's contract simply becomes more valuable.

Think:

> "I have a valuable contract, but I haven't received the money yet."

The gain is:

$$\boxed{\text{Unrealized MTM gain}}$$

Step 3: Ace's futures contract

Now look at Ace.

Ace has a **long futures contract**.

XLYA rises.

Therefore:

$$\boxed{\text{Ace gains on the futures}}$$

Approximately:

$$€15 \times 100 = \boxed{€1,500}$$

But futures are different.

Futures are:

$$\boxed{\text{marked to market daily}}$$

Therefore, Ace's €1,500 gain is **settled today**.

The exchange/clearing system deposits the gain into Ace's futures margin account.

So:

$$\boxed{\text{Ace receives approximately €1,500 today}}$$

Step 4: Put the two contracts side by side

This is the entire question:

	Client's Forward	Ace's Futures
Position	Long	Long
XLYA rises	Gain	Gain
MTM changes?	Yes	Yes
Daily cash settlement?	❌ No	✅ Yes
Cash today?	❌ No	✅ Approximately €1,500
Where?	—	Ace's margin account

That's exactly what answer **B** is saying.

Step 5: Why does Ace need this futures gain?

This is the **hedging** part.

Remember:

The client is long the forward.

Therefore, if the stock rises, the client's forward becomes more valuable.

But from **Ace's perspective**, that's a liability.

Client:

$$\boxed{\text{Long forward} \rightarrow \text{gain}}$$

Ace:

$$\boxed{\text{Short side of forward} \rightarrow \text{loss}}$$

So:

$$\text{XLYA rises}$$

$$\downarrow$$

Client gains on forward

$$\downarrow$$

Ace loses on forward

But Ace has hedged with a **long futures**:

$$\text{XLYA rises}$$

$$\downarrow$$

Ace gains on futures

$$\downarrow$$

$$\boxed{\text{Futures gain offsets forward loss}}$$

That's the whole purpose of the hedge.

Step 6: But the cash flows happen at different times

This is the subtle CFA point.

Suppose Ace loses:

$$€1,500$$

economically on the forward.

But Ace gains:

$$€1,500$$

on the futures.

You might think:

> "Then they just cancel and nothing happens."

Not exactly.

The **economic exposure** is hedged, but the **cash-flow timing** is different.

Forward:

$$\boxed{\text{Gain/loss recognized in value, but no daily cash}}$$

Futures:

$$\boxed{\text{Gain/loss settled daily in cash}}$$

So Ace receives cash from the futures margin account **today**.

Step 7: A simple everyday analogy

Imagine you owe your friend $1,500 six months from now.

Today, your friend tells you:

> "Your debt has become $1,500 more expensive."

That's like the **forward MTM**.

You haven't paid the $1,500 today.

Now imagine you have another investment that gives you $1,500 **today** because the market moved in your favor.

That's like the **futures MTM**.

Economically, the two can offset:

$$\boxed{-\$1,500 + \$1,500 \approx 0}$$

But the **timing of the cash flows is different**.

Step 8: Why the answer says "if spot and futures prices change by approximately the same amount"

This is another subtle point.

The stock's spot price increased:

$$€15$$

But the futures price doesn't necessarily increase by exactly €15.

Why?

Because futures prices also depend on:

- interest rates
- time to maturity
- carrying costs
- basis

But over a short period, the futures and spot prices can move approximately together.

So the question says:

> **if spot and futures prices change by approximately the same amount**

then:

$$€15 \times 100 = \boxed{€1,500}$$

is approximately the futures MTM gain.

⭐ The most important distinction

Please memorize this:

FORWARD

$$\boxed{\text{MTM changes}}$$

but:

$$\boxed{\text{No daily cash settlement}}$$

FUTURES

$$\boxed{\text{MTM changes}}$$

and:

$$\boxed{\text{Daily cash settlement}}$$

🧠 CFA exam shortcut

Whenever you see a question comparing **forward vs. futures**, immediately ask:

Question 1:

Does the value change?

Both:

$$\boxed{\text{Forward AND futures}}$$

Yes.

Question 2:

Is the gain/loss paid in cash every day?

Forward:

$$\boxed{\text{NO}}$$

Futures:

$$\boxed{\text{YES}}$$

That's the entire concept behind this question.

Final picture



```
        ▼                ▼
  Client gains      Ace gains
  MTM ≈ €1,500      MTM ≈ €1,500
        |                |
        ▼                ▼
  NO CASH TODAY     CASH TODAY
                    into margin
                    account
```

Therefore:

B is correct

One sentence to remember:

> **A forward's MTM gain is generally unrealized until maturity, whereas a futures MTM gain is settled in cash every day through the margin account.**

Identify which of the following corresponds to which description.
1. Long interest rate futures position A. Results in a gain when MRR settles above
the initial forward commitment rate at
maturity
2. Pay fixed (receive floating) FRA contract B. Results in a loss when MRR settles above
the initial forward commitment rate at
maturity
3. Receive fixed (pay floating) FRA contract C. Has a

Show more

Yes. This is a **matching question**, and the easiest way is to understand what each position wants interest rates to do.

Step 1: Match the FRA positions first

2. Pay fixed / receive floating FRA

You:

- **Pay fixed rate**
- **Receive MRR (floating rate)**

Suppose:

$$\text{FRA fixed rate} = 4\%$$

If MRR rises to:

$$5\%$$

you receive 5% but pay 4%.

So:

$$5\% - 4\% = +1\%$$

Therefore:

$$\boxed{\text{Pay fixed / receive floating} \rightarrow \text{gain when MRR rises}}$$

So:

$$\boxed{2 \rightarrow A}$$

---

3. Receive fixed / pay floating FRA

You:

- **Receive fixed**
- **Pay MRR**

Suppose:

$$\text{Fixed} = 4\%$$

and MRR rises to:

$$5\%$$

You receive 4%, but have to pay 5%.

Therefore:

$$4\% - 5\% = -1\%$$

So:

$$\boxed{\text{Receive fixed / pay floating} \rightarrow \text{loss when MRR rises}}$$

Therefore:

$$\boxed{3 \rightarrow B}$$

Step 2: Now deal with the interest-rate futures

We have left:

$$\boxed{1.\ Long\ interest\ rate\ futures}$$

and:

$$\boxed{C.\ Has\ a\ forward\ commitment\ price\ that\ will\ increase\ as\ short-term\ in}$$

So:

$$\boxed{1 \to C}$$

Why does a long interest-rate futures position correspond to C?

This is where the **100 − yield** convention matters.

Interest-rate futures are commonly quoted as:

$$\boxed{\text{Futures price} = 100 - \text{implied interest rate}}$$

For example:

If MRR/yield = 5%

$$100 - 5 = 95$$

Futures price:

$$95$$

If interest rate falls to 4%

$$100 - 4 = 96$$

Futures price:

$$96$$

So:

$$\boxed{\text{Interest rates } \downarrow \Rightarrow \text{futures price } \uparrow}$$

Therefore, a **long interest-rate futures position** benefits when interest rates fall.

That's why:

$$\boxed{1 \rightarrow C}$$

Final answer

Item	Position	Correct description
1	Long interest-rate futures	**C**
2	Pay fixed / receive floating FRA	**A**
3	Receive fixed / pay floating FRA	**B**

So:

$$\boxed{\mathbf{1-C},\quad \mathbf{2-A},\quad \mathbf{3-B}}$$

🧠 The easiest memory trick

Think of the three positions as follows:

Long interest-rate futures

$$\boxed{\text{Rates} \downarrow \rightarrow \text{Futures price} \uparrow \rightarrow \text{Gain}}$$

Pay fixed / receive floating FRA

$$\boxed{\text{Rates} \uparrow \rightarrow \text{Floating received} \uparrow \rightarrow \text{Gain}}$$

Receive fixed / pay floating FRA

$$\boxed{\text{Rates} \uparrow \rightarrow \text{Floating paid} \uparrow \rightarrow \text{Loss}}$$

So you can memorize:

> **Long futures likes FALLING rates.**
> **Pay-fixed FRA likes RISING rates.**
> **Receive-fixed FRA likes FALLING rates.**

That last sentence is particularly useful for CFA exam questions.

ChatGPT can make mistakes. Check important info.